

Network Functions Virtualization



What exactly is Network Functions Virtualization

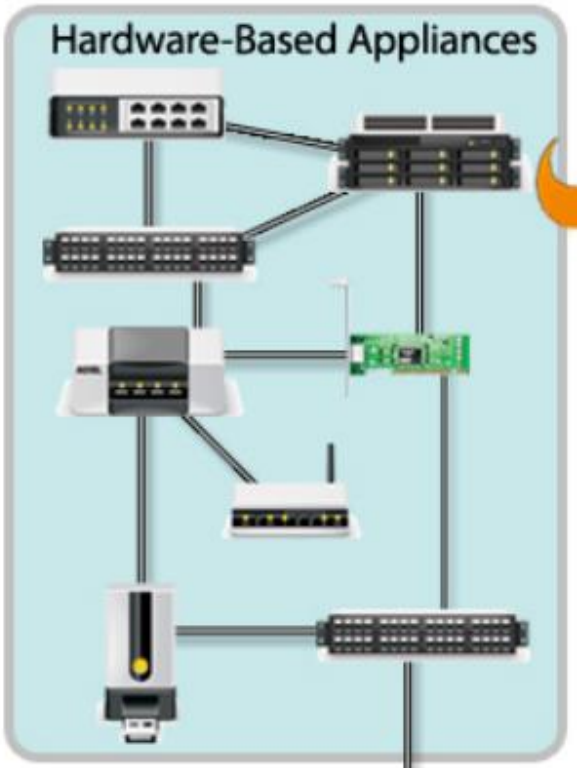
- Network Function Virtualization, or NFV, is a way to reduce cost and accelerate service deployment for network operators by decoupling functions like a firewall or encryption from dedicated hardware and moving them to virtual servers.

- Customer wants to add a new network function, the service provider can simply spin up a new virtual machine to perform that function.
- Virtualization of network functions reduces dependency on dedicated hardware appliances for network operators, and allows for improved scalability and customization across the entire network.
- Different from a virtualized network, NFV seeks to offload network functions only, rather than the entire network.



Classic Network Appliance Approach

Hardware-Based Appliances





Fragmented, Non-Standard Hardware

NFV Approach



NFV enables virtualized network functions to run over an open hardware platform, reducing CapEx, OpEx, and accelerating innovation.



High Volume, Standard Server



High Volume, Standard Storage



High Volume, Standard Switch



Cloud Computing ?







What are the advantages of NFV?

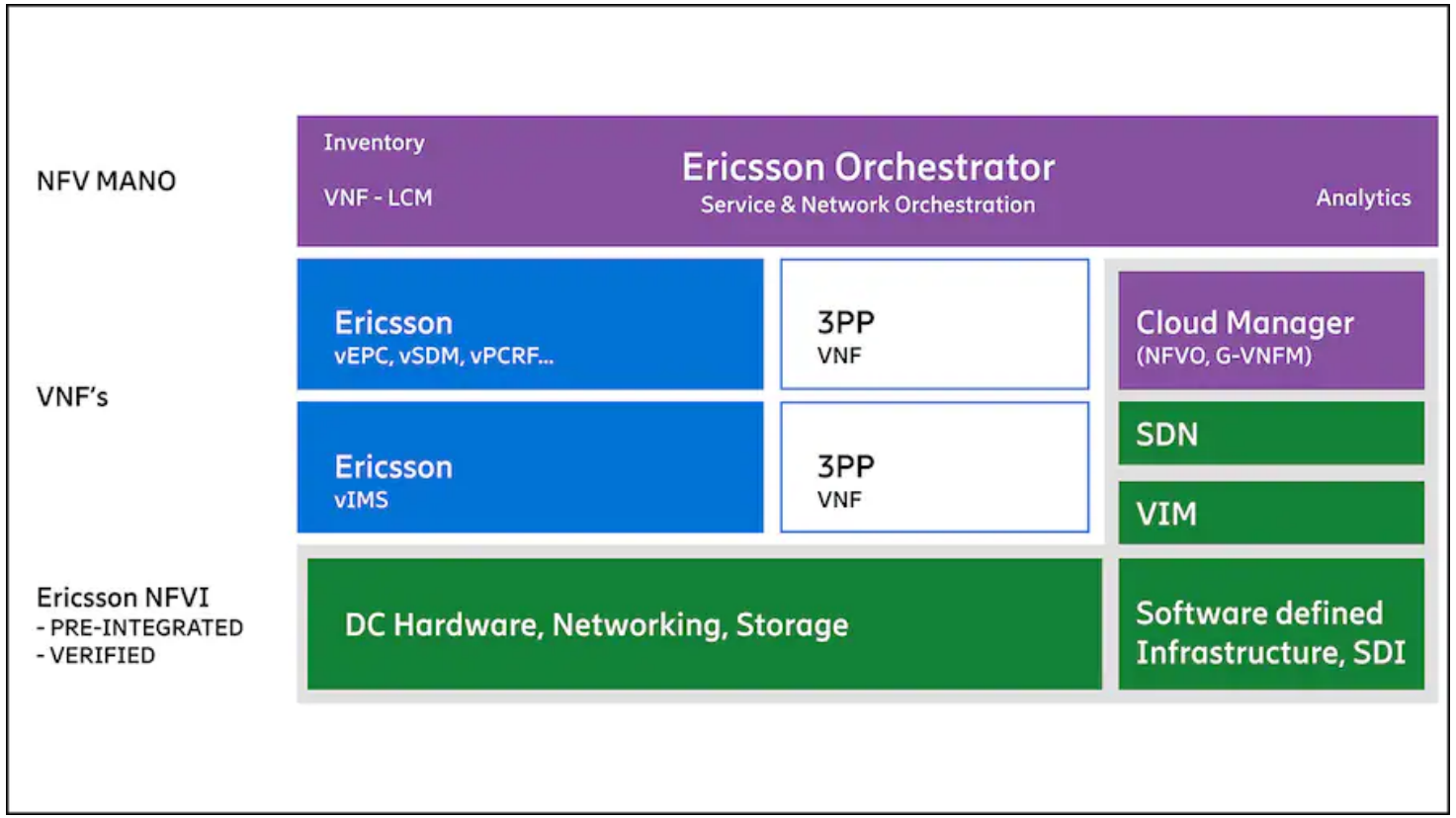
NFV reduces the need for dedicated hardware to deploy and manage networks

Can be managed from anywhere within the operator's network.

- Reduced space needed for network hardware
- Reduce network power consumption
- Reduced network maintenance costs
- Easier network upgrades
- Longer life cycles for network hardware
- Reduced maintenance and hardware costs



NFV Usage in the world





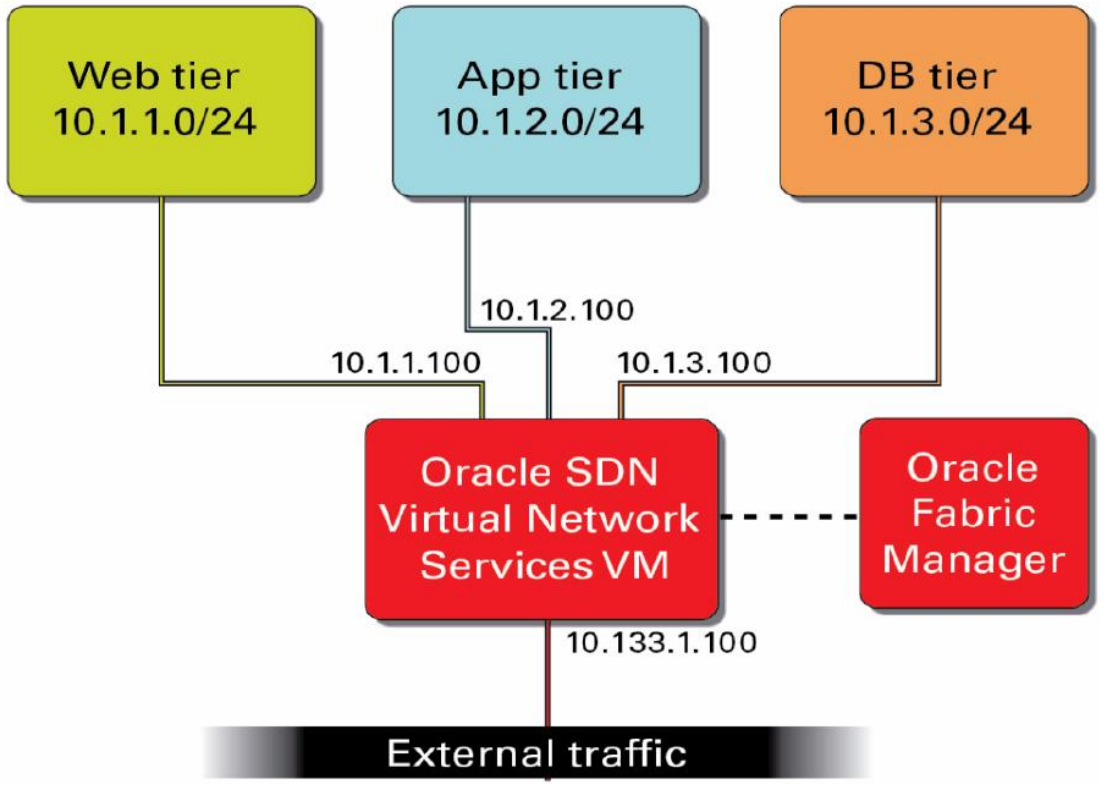
NFV Usage in the world

- NFV allows network operators to manage and expand their network capabilities on demand using virtual, software based applications where physical boxes once stood in the network architecture.

- This makes it easier to load-balance, scale up and down, and move functions across distributed hardware resources. With continual updates, operators can keep things running on the latest software without interruption to their customers.

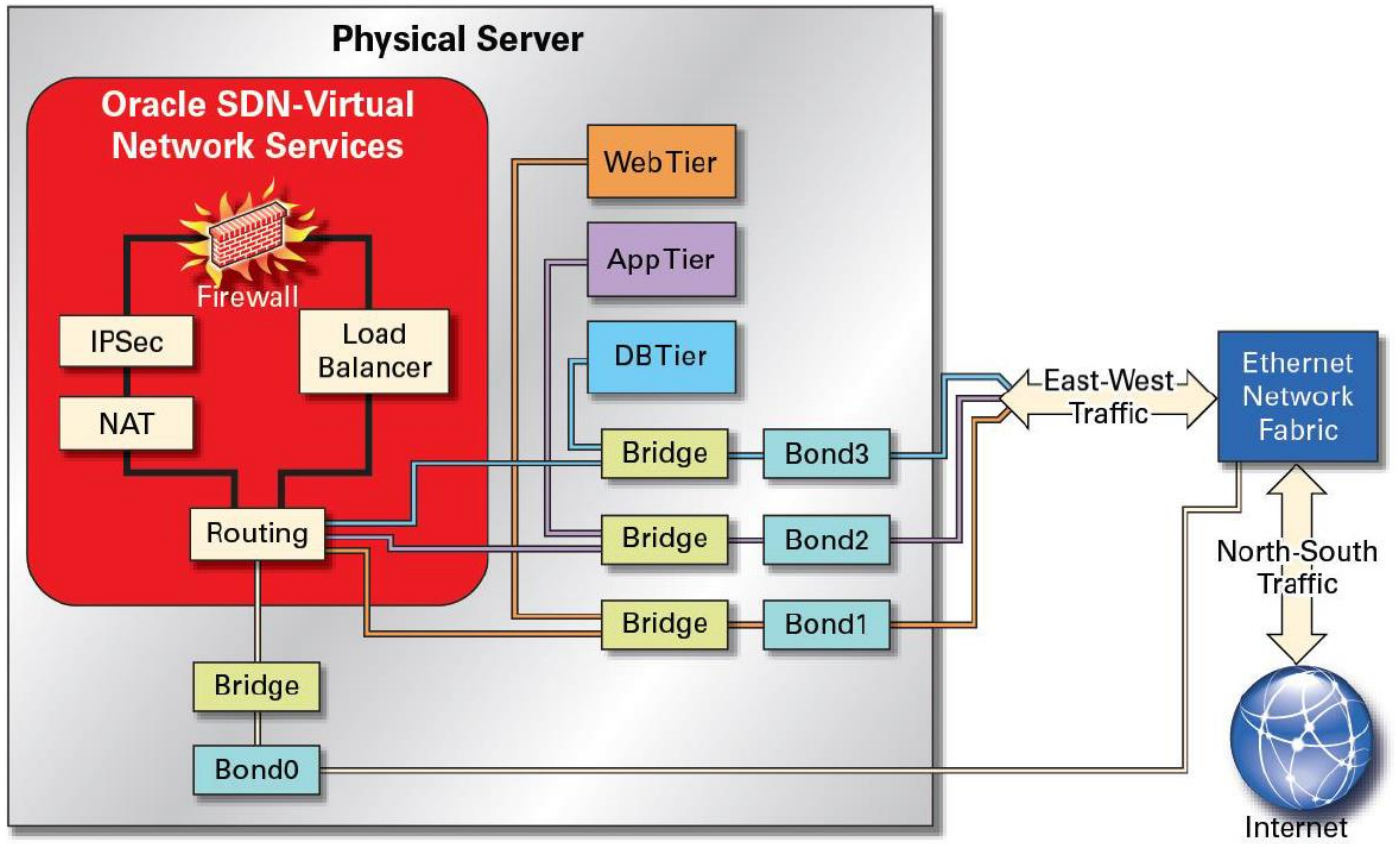


NFV Usage in the world-Oracle



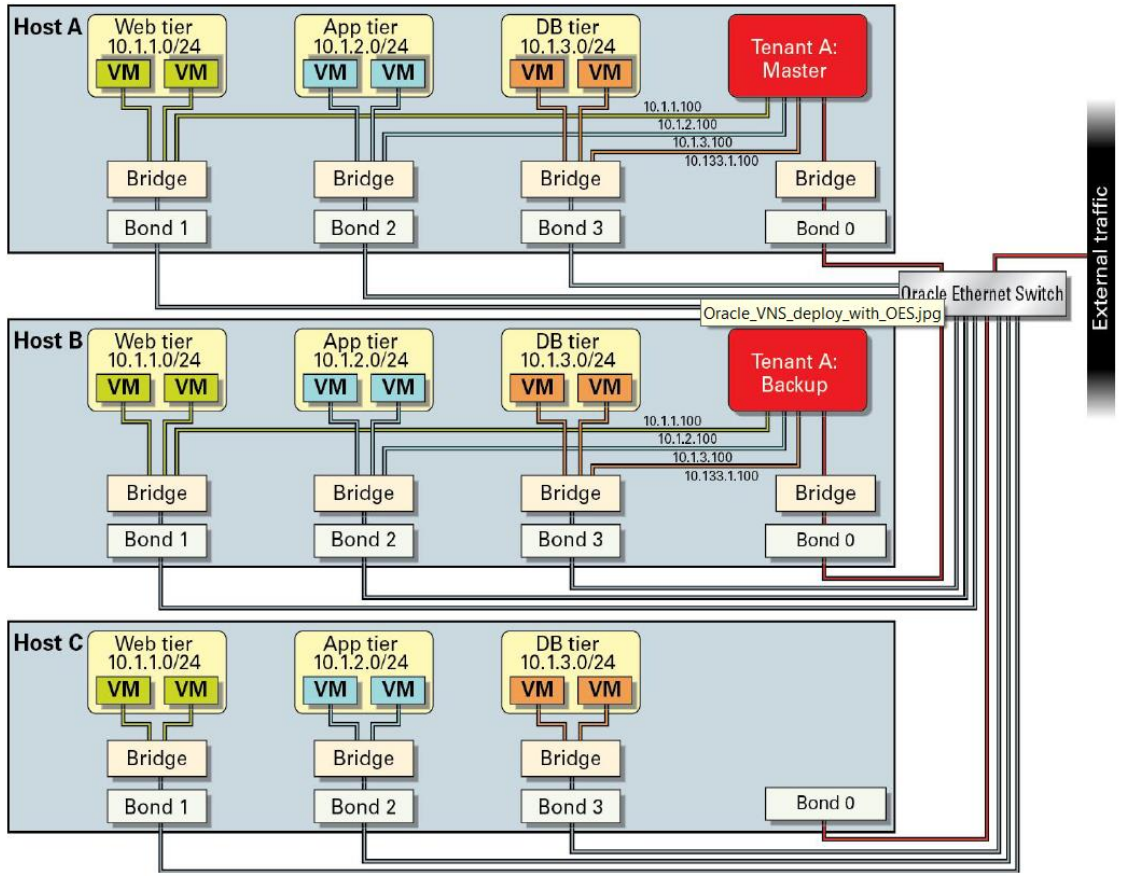


SDN Industry Usage



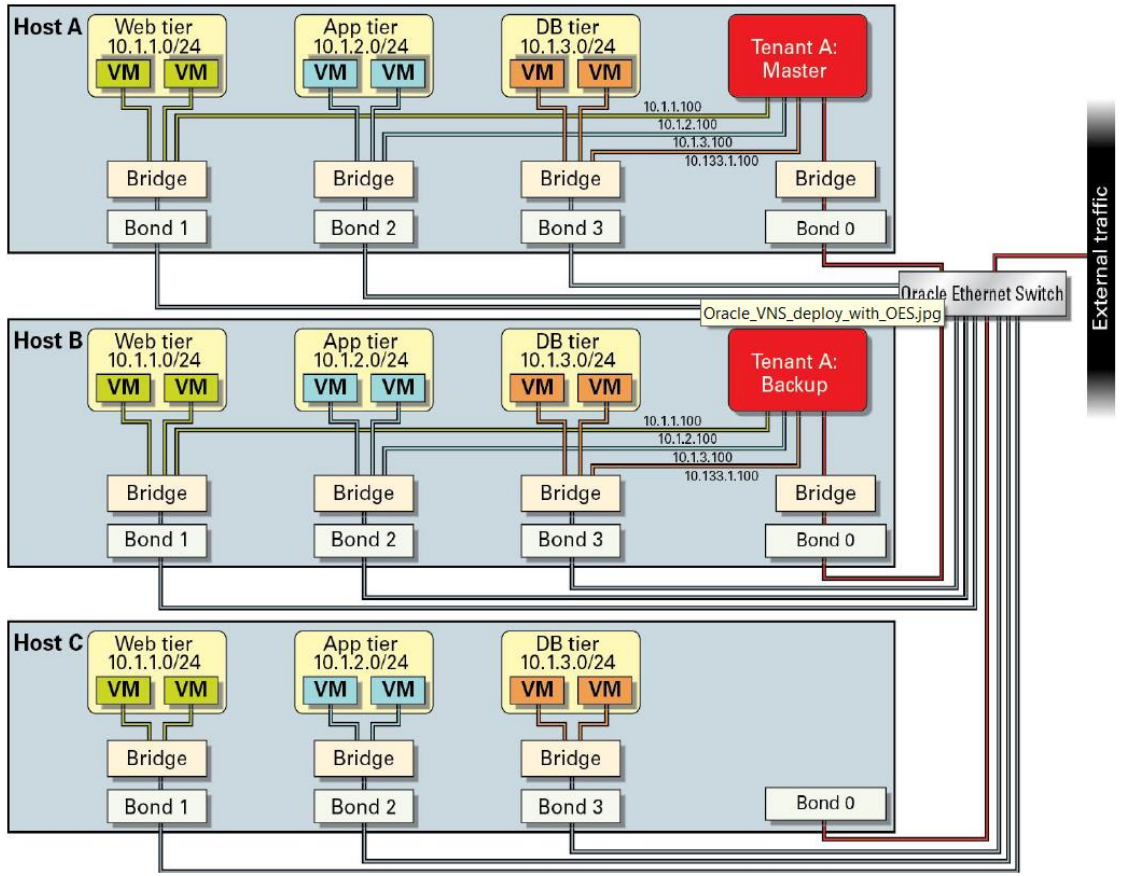


SDN Industry Usage Detail





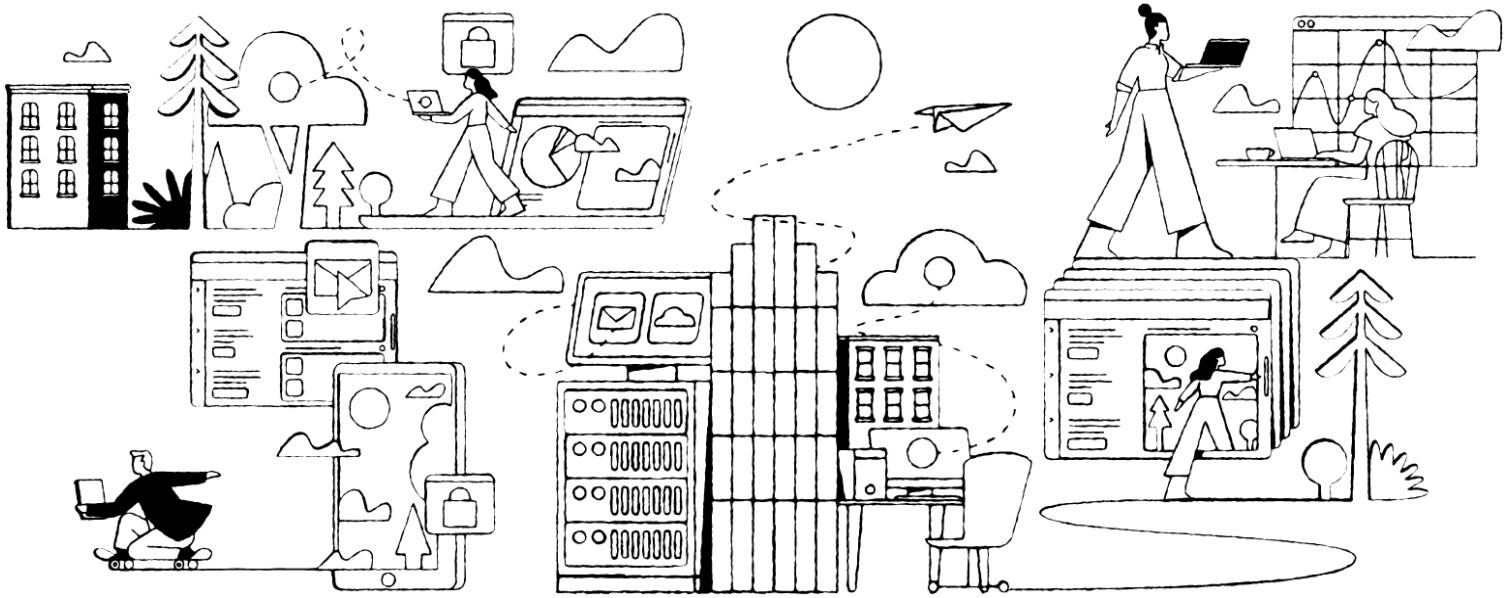
SDN Industry Usage Detail




Evolution of NFV
as the platform for
Strategic Product Manager with Cloud and SDN.



Citrix DAAS



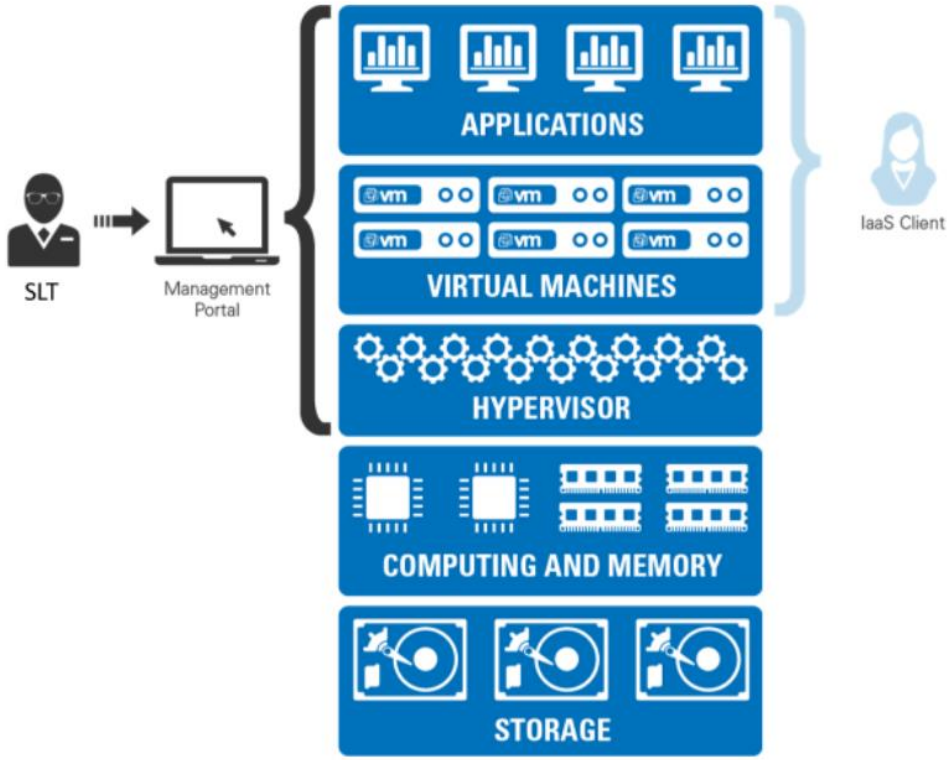


Citrix Usage on the world ?

- Citrix Virtual Apps and Desktops in place in our on-premises datacenter.
- Reduce lengthy hardware procurement cycle
- Citrix DaaS provides advanced monitoring and cloud-native management—everything you need to scale fast and meet the needs of your flexible workforce

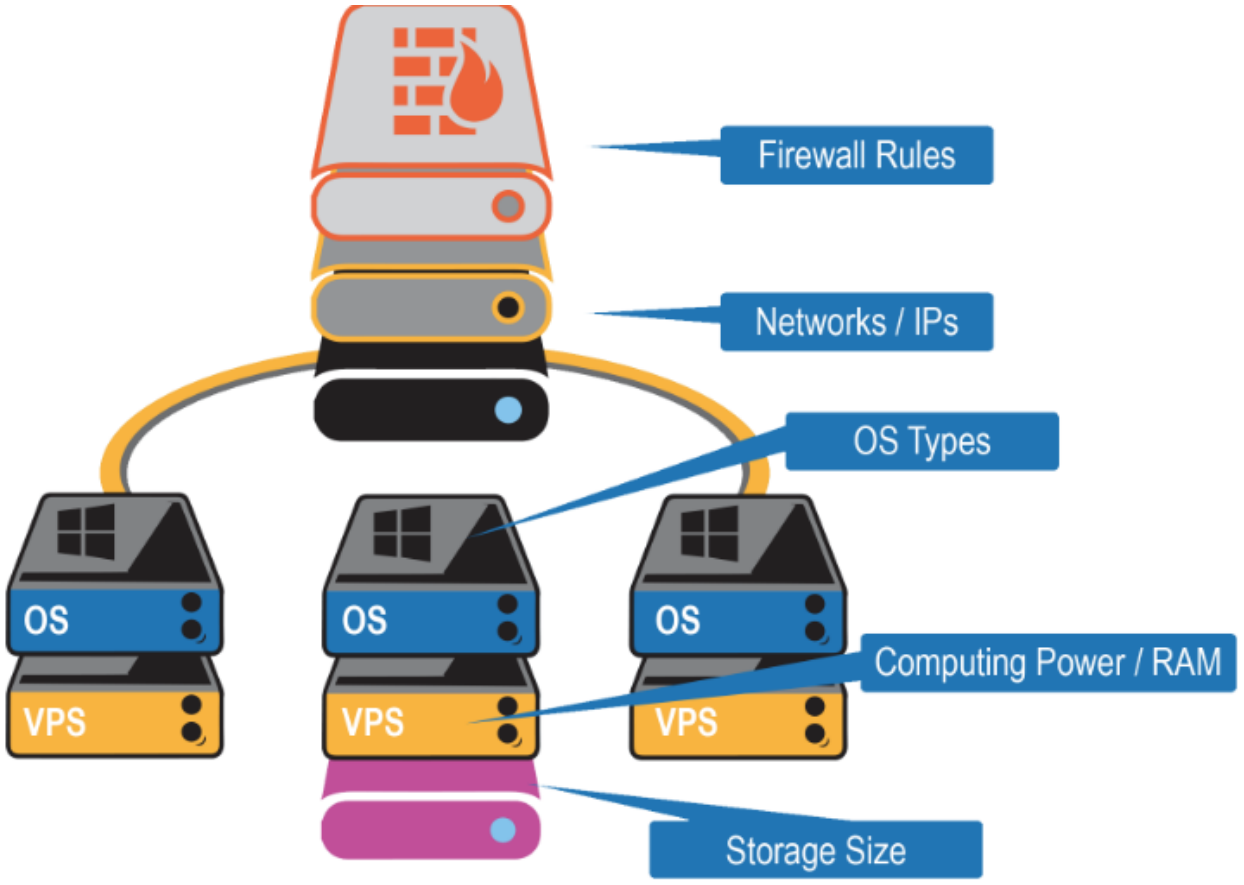


Akasa Cloud


SLT
Management Portal
APPLICATIONS
VIRTUAL MACHINES
HYPERVISOR
COMPUTING AND MEMORY
STORAGE
IaaS Client



Akasa Cloud Continued

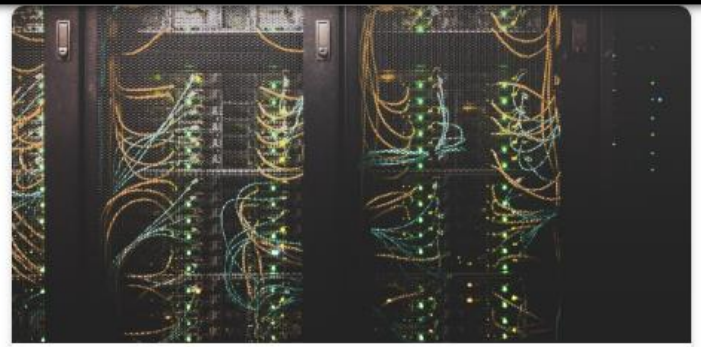


Dialog Cloud



Desktop as a Service (DaaS)

Desktop as a Service (DaaS) is a form of virtual desktop infrastructure service (VDI) that is hosted...



Dialog Cloud Backup as a Service (BaaS)

Dialog Cloud Backup-as-a-Service (BaaS) is a data protection solution that leverages the Dialog Cloud to provide tapeless...



Dialog Co-Location

Sri Lanka's Most Trusted Data Centre Provider Dialog Data Centre (DC) enables critical co-location, connectivity and cloud...

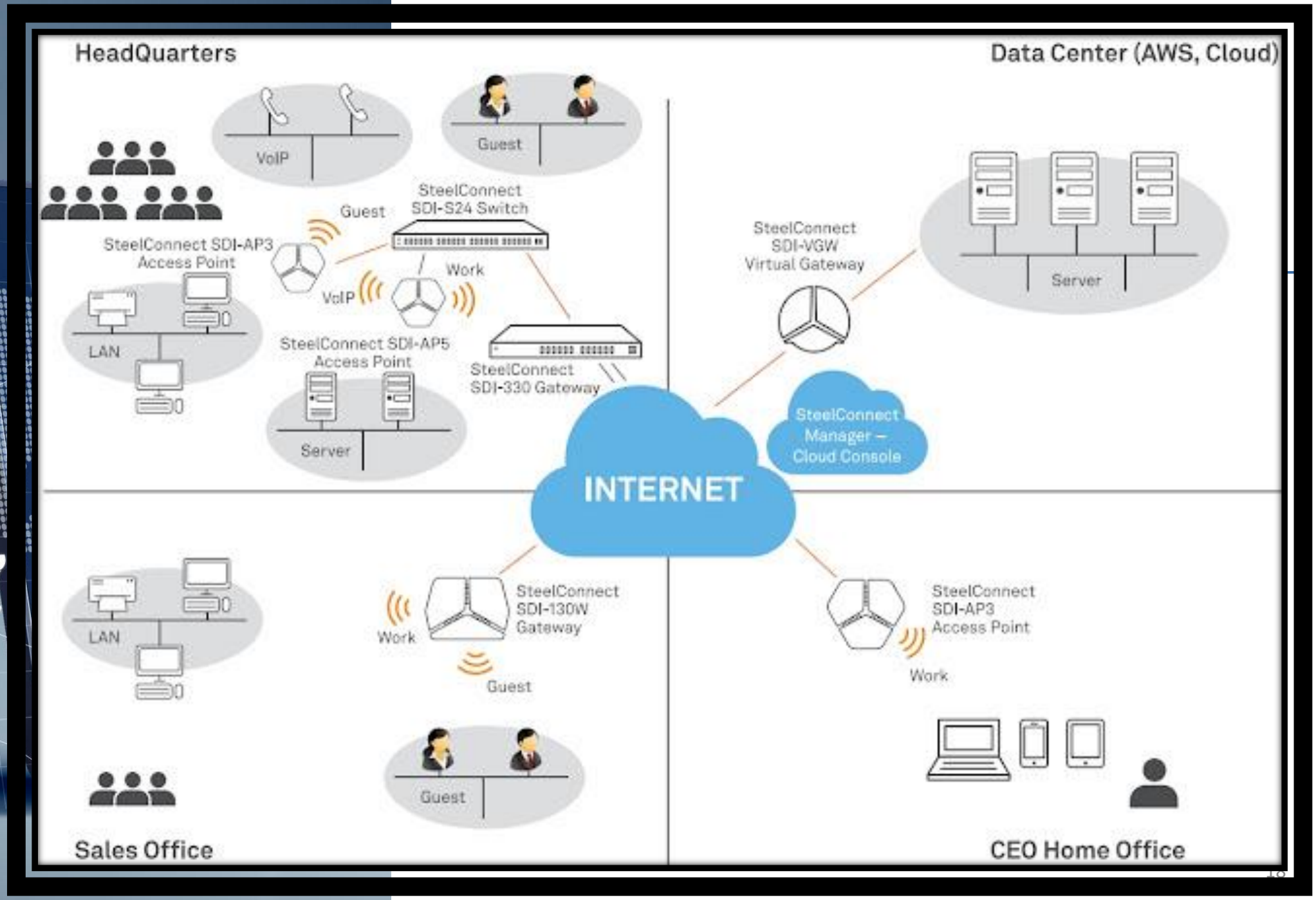


Dialog Cloud Disaster Recovery as a Service (DRaaS)



Dialog Enterprise Cloud

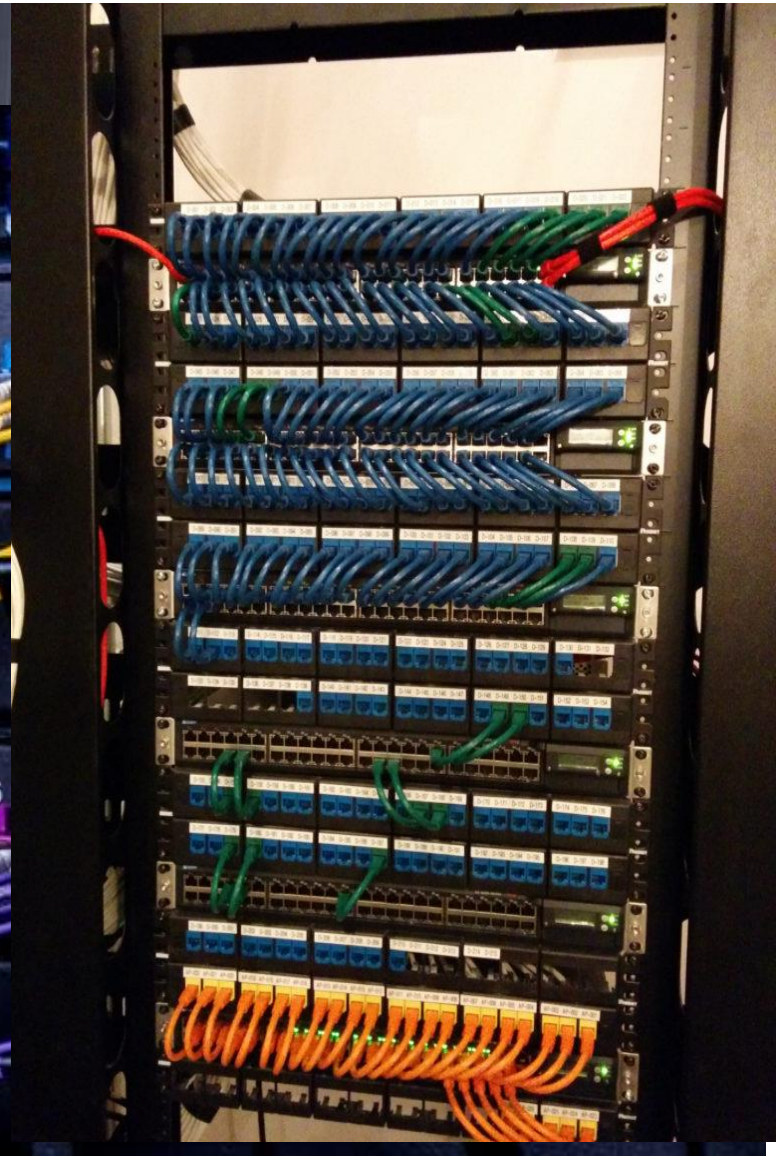
Extending the capabilities of a Software-Defined


HeadQuarters
Data Center (AWS, Cloud)
VoIP
Guest
SteelConnect
SDI-S24 Switch
Guest
SteelConnect SDI-AP3
Access Point
Work
VoIP
LAN
SteelConnect SDI-AP5
Access Point
SteelConnect
SDI-330 Gateway
Server
SteelConnect
SDI-VGW
Virtual Gateway
Server
SteelConnect
Manager –
Cloud Console
INTERNET
LAN
Work
SteelConnect
SDI-130W
Gateway
Guest
Guest
SteelConnect
SDI-AP3
Access Point
Work
Sales Office
CEO Home Office

Wired Media

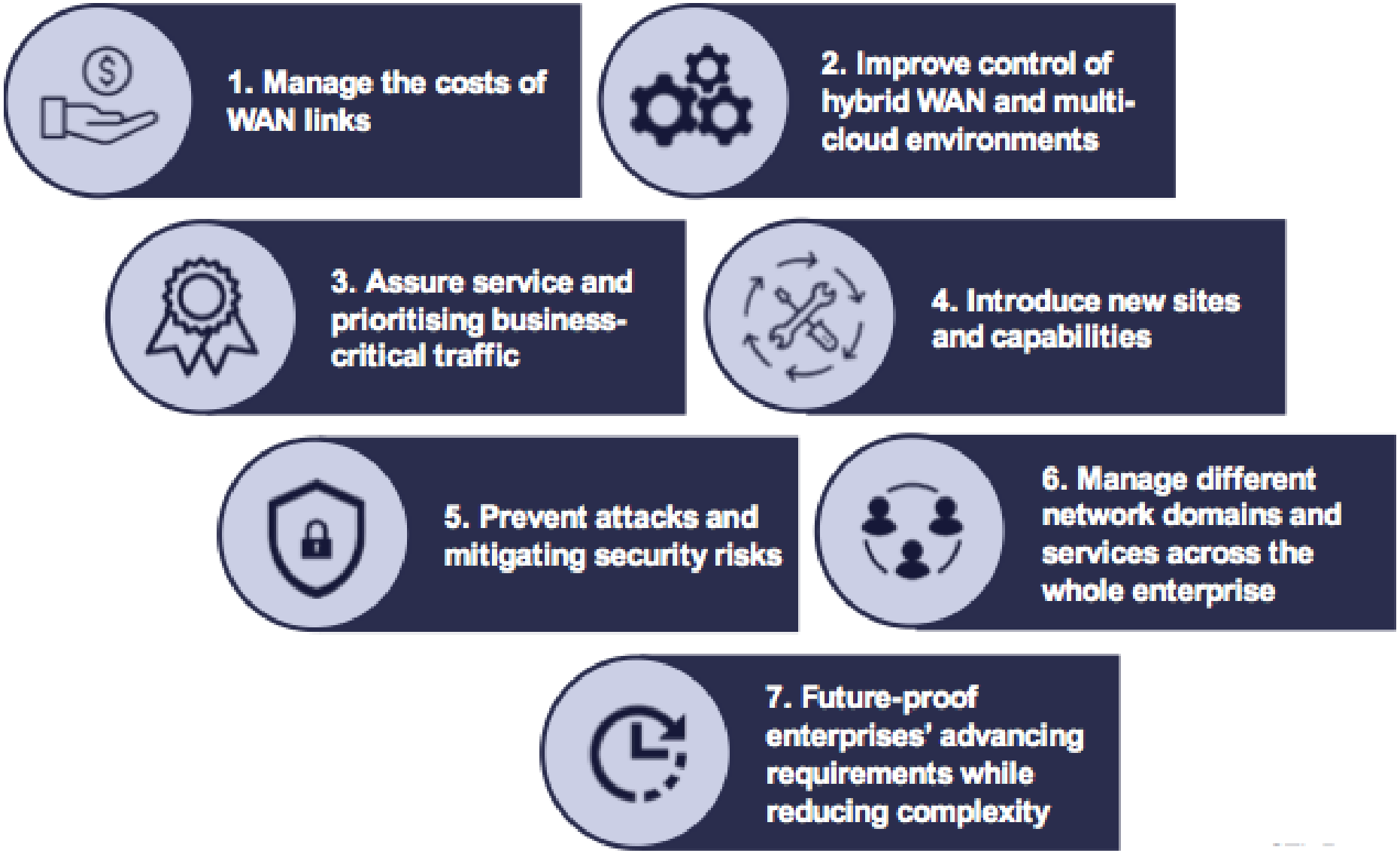
1. Structured Caballing Systems



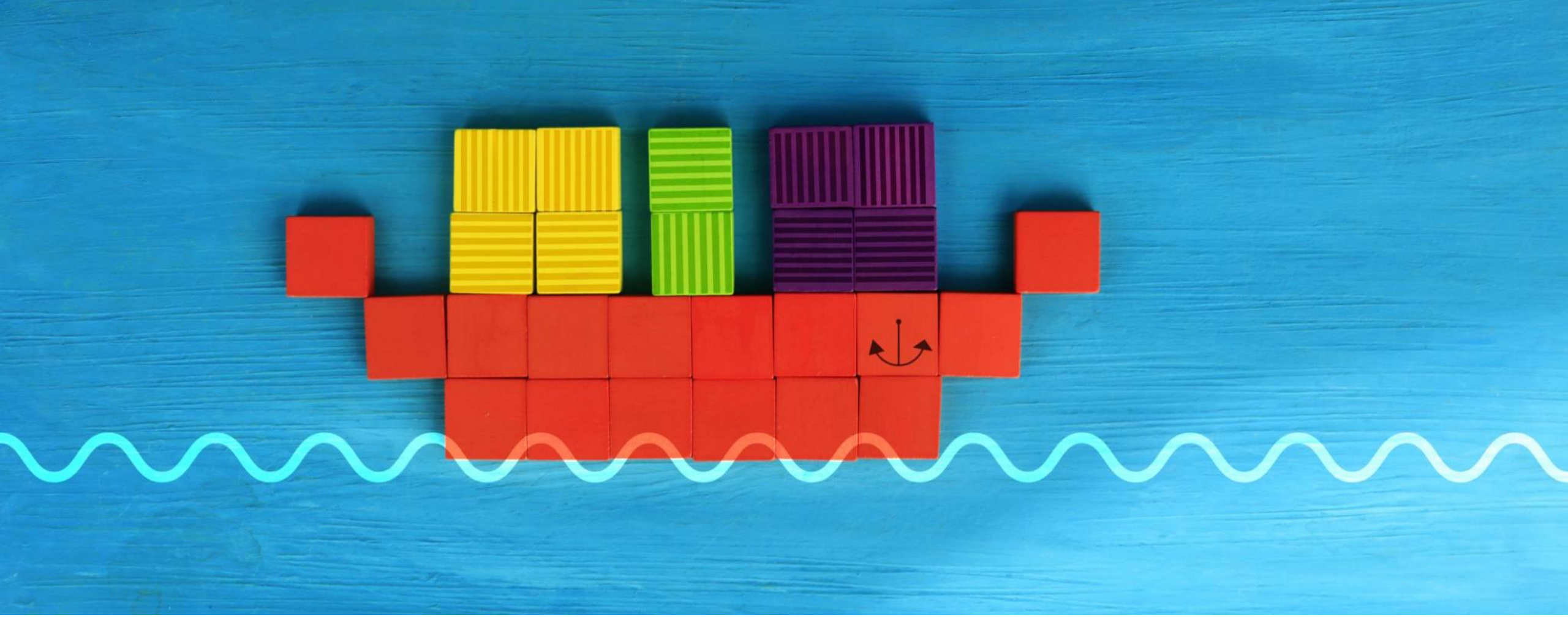



0:00
-1:42

WHY SDN?

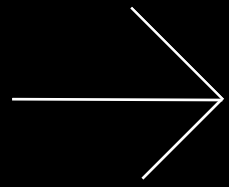

1. Manage the costs of WAN links
2. Improve control of hybrid WAN and multi-cloud environments
3. Assure service and prioritising business-critical traffic
4. Introduce new sites and capabilities
5. Prevent attacks and mitigating security risks
6. Manage different network domains and services across the whole enterprise
7. Future-proof enterprises' advancing requirements while reducing complexity

Containers



Docker and Containers

Foundations of Containers





Motivation for Containers and Modern Application Deployment

Challenges of Traditional Deployment

Traditional deployment faced issues like dependency conflicts and inconsistent environments causing 'works on my machine' problems.

Container Packaging Benefits

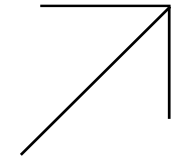
Containers bundle applications with all dependencies ensuring consistent behavior across development, testing, and production.

Support for Microservices Architecture

Containers enable efficient deployment and scaling of independent microservices on shared hosts with speed and flexibility.

Improved Software Delivery

Containers reduce environment-related errors and speed up development and deployment cycles for modern applications.



Virtualization and Container Technology

Virtual Machines vs Containers: Architectural Differences



Virtual Machines Architecture

Virtual machines run full guest operating systems on a hypervisor, providing strong isolation but with resource overhead.

Containers Architecture

Containers share the host OS kernel using namespaces and cgroups, enabling lightweight and fast startup.

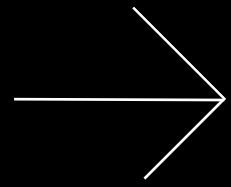
Resource Efficiency and Density

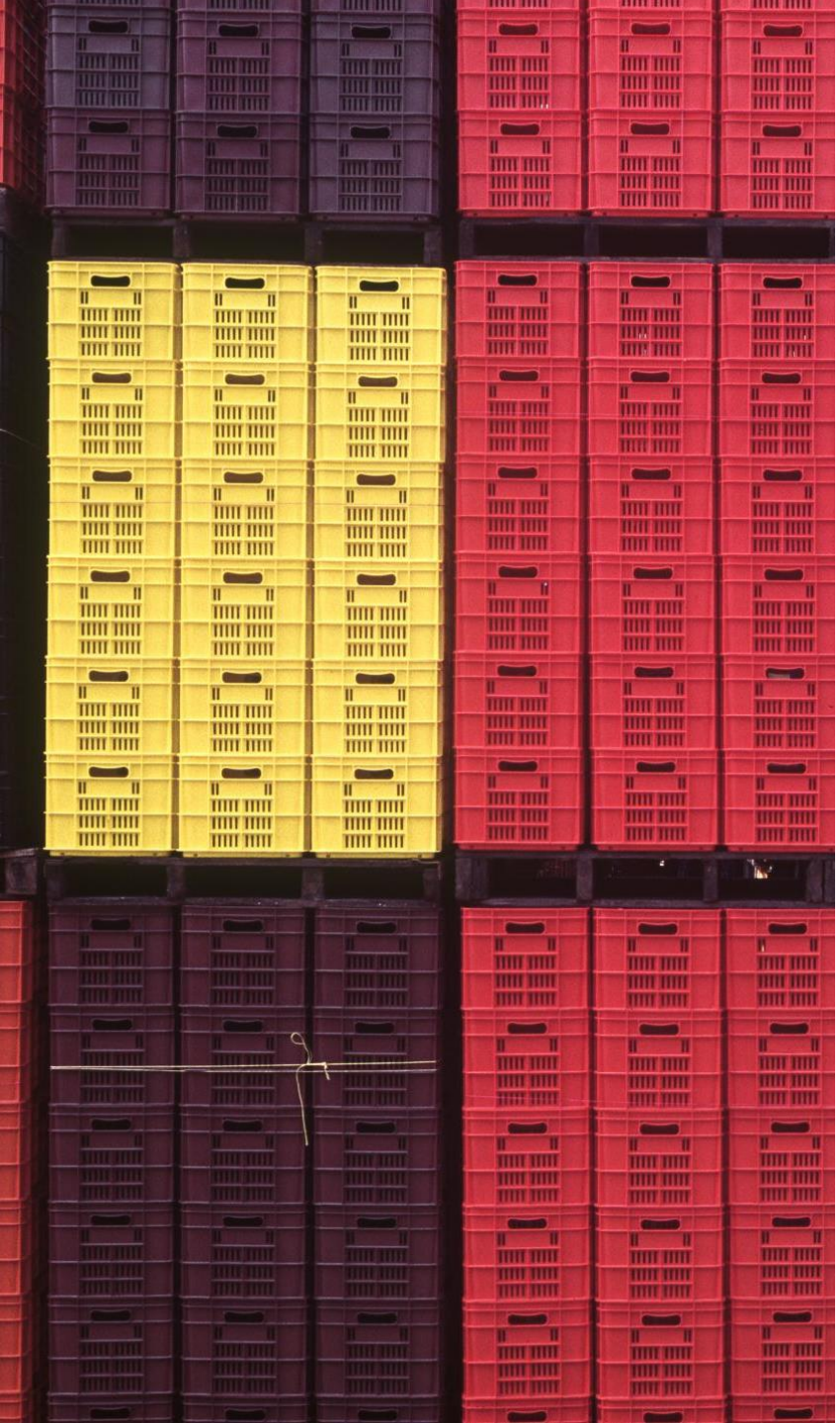
Containers enable higher density and more efficient resource usage compared to virtual machines on the same hardware.

Use Case Suitability

Containers suit fast deployment and portability, while VMs suit strict isolation and diverse OS needs.

Docker Platform Overview





Docker Architecture and Core Components

Client-Server Architecture

Docker uses a client-server model where the client sends commands to the Docker daemon managing containers.

Docker Daemon Functions

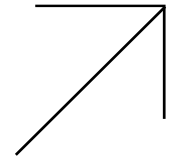
The Docker daemon builds images, runs containers, manages networks, volumes, and communicates with registries.

Docker Images and Containers

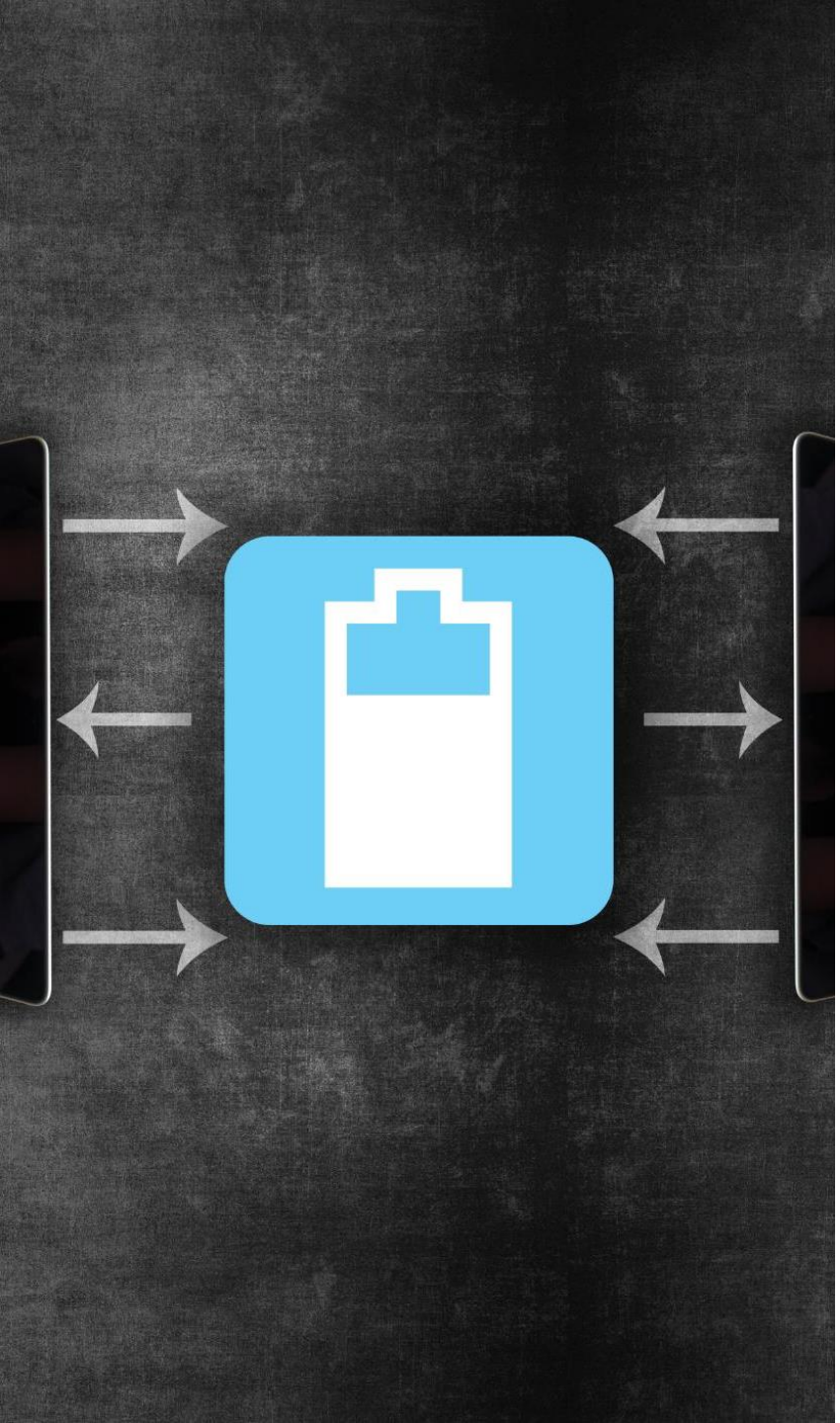
Docker images are read-only templates; containers are running instances created from images.

Image Registries

Registries provide centralized storage and sharing for Docker images, including public and private repositories.



Building and Running Containers



Docker Images, Dockerfiles, and the Container Lifecycle

Docker Image Creation

Docker images are built from Dockerfiles containing step-by-step instructions specifying the image's configuration and contents.

Layered Image Build Process

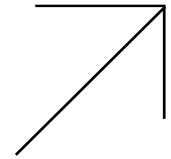
Each Dockerfile instruction creates a new image layer, enabling caching and efficient incremental builds.

Container Lifecycle

Containers go through creation, start, stop, restart, and removal stages and are designed to be ephemeral.

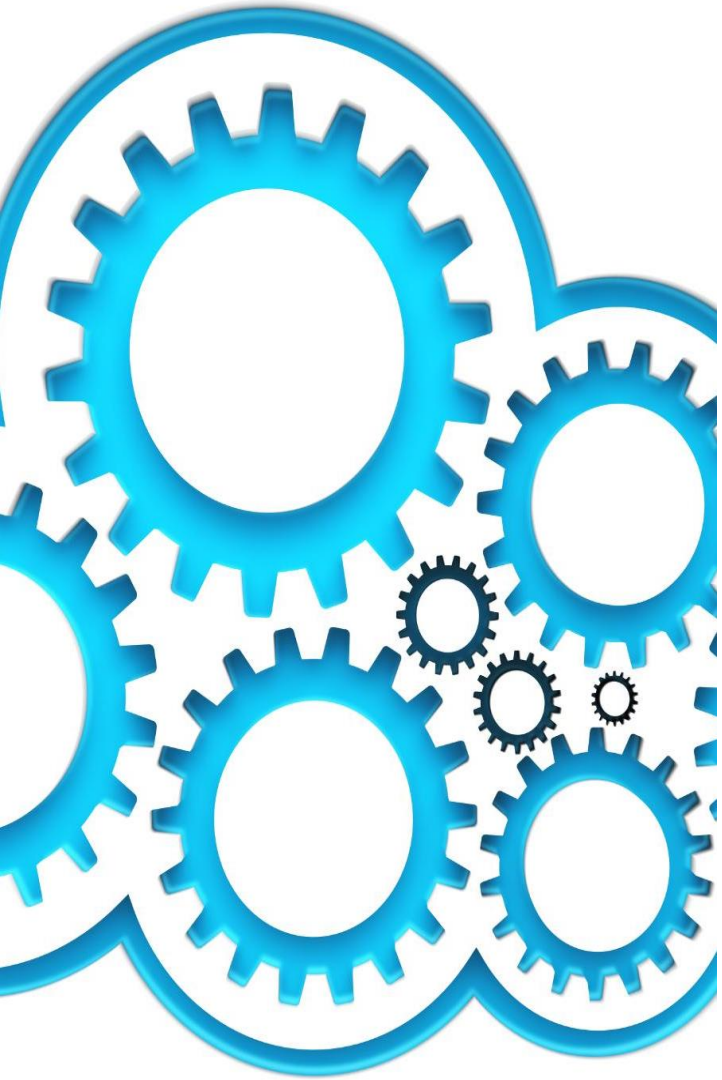
Persistent Data Management

Persistent data is stored outside containers using volumes to prevent data loss during container removal.



Docker in Practice and Industry Context

Use Cases, Benefits, and Limitations of Docker



Application Packaging

Docker bundles applications with their dependencies, enabling consistent environments for development and testing.

Production Deployment Benefits

Docker allows faster, reliable deployments with support for rolling updates and horizontal scaling of services.

Security Limitations

Containers share the host OS kernel, requiring careful security practices to mitigate risks compared to virtual machines.

Efficiency and Portability

Docker enhances software engineering by enabling portability, efficiency, and faster release cycles across environments.

Overview

- AWS Global Infrastructure
- Disaster recovery strategies
- Recovery point of objectives
- Maintaining Business Continuity
- Route 53 routing policies
- AWS migration services



AWS Global Infrastructure

Regions



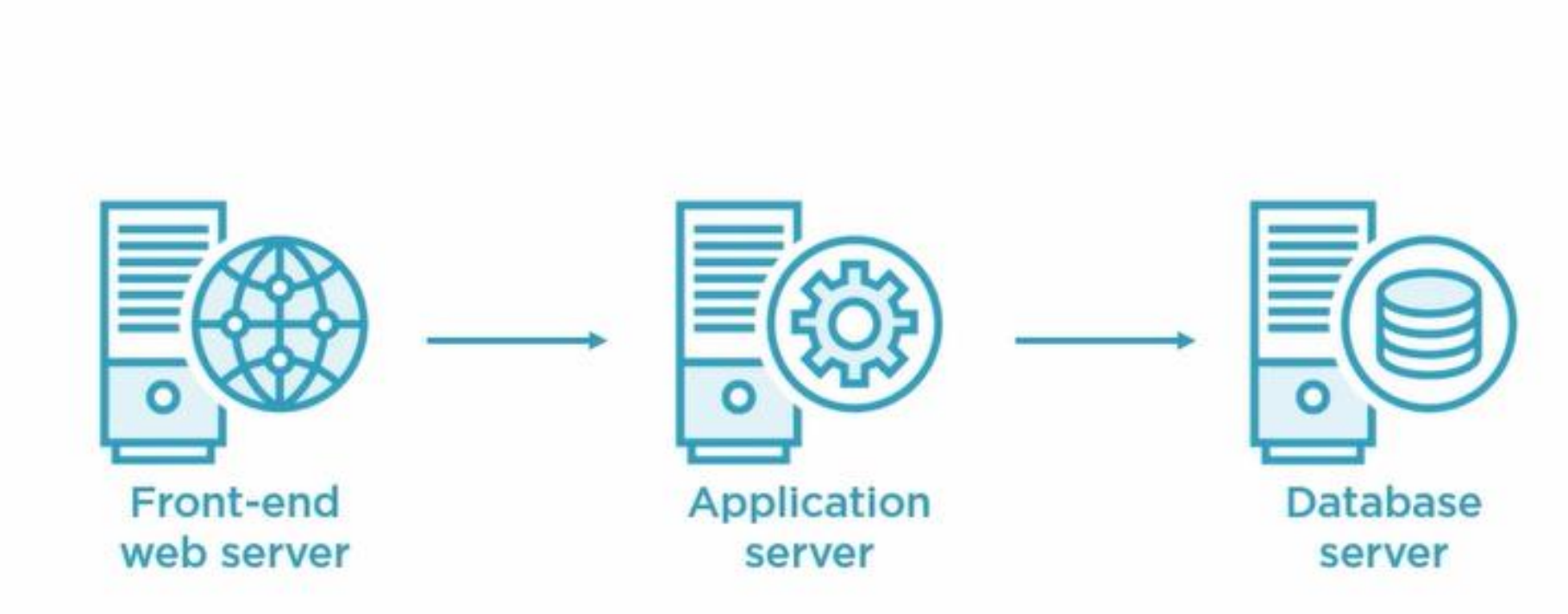
Availability Zones



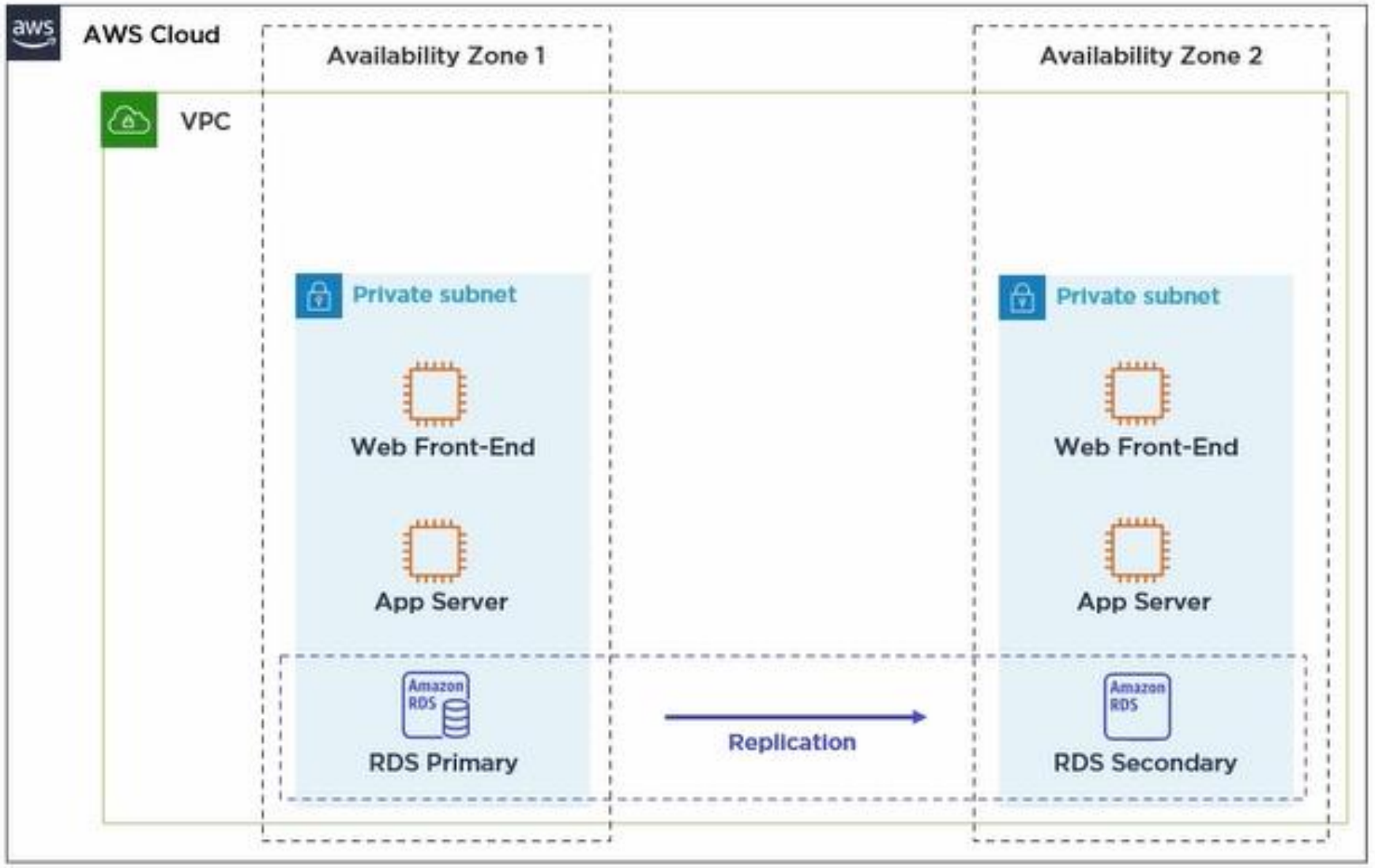
- Located all around the world
- Servers and services are based in regions
- Choose based on user locations
- Cluster of data centers
- Independent power and cooling
- Multiple redundant network connections
- All regions have at least 2 AZ

- Single datacenter experiences

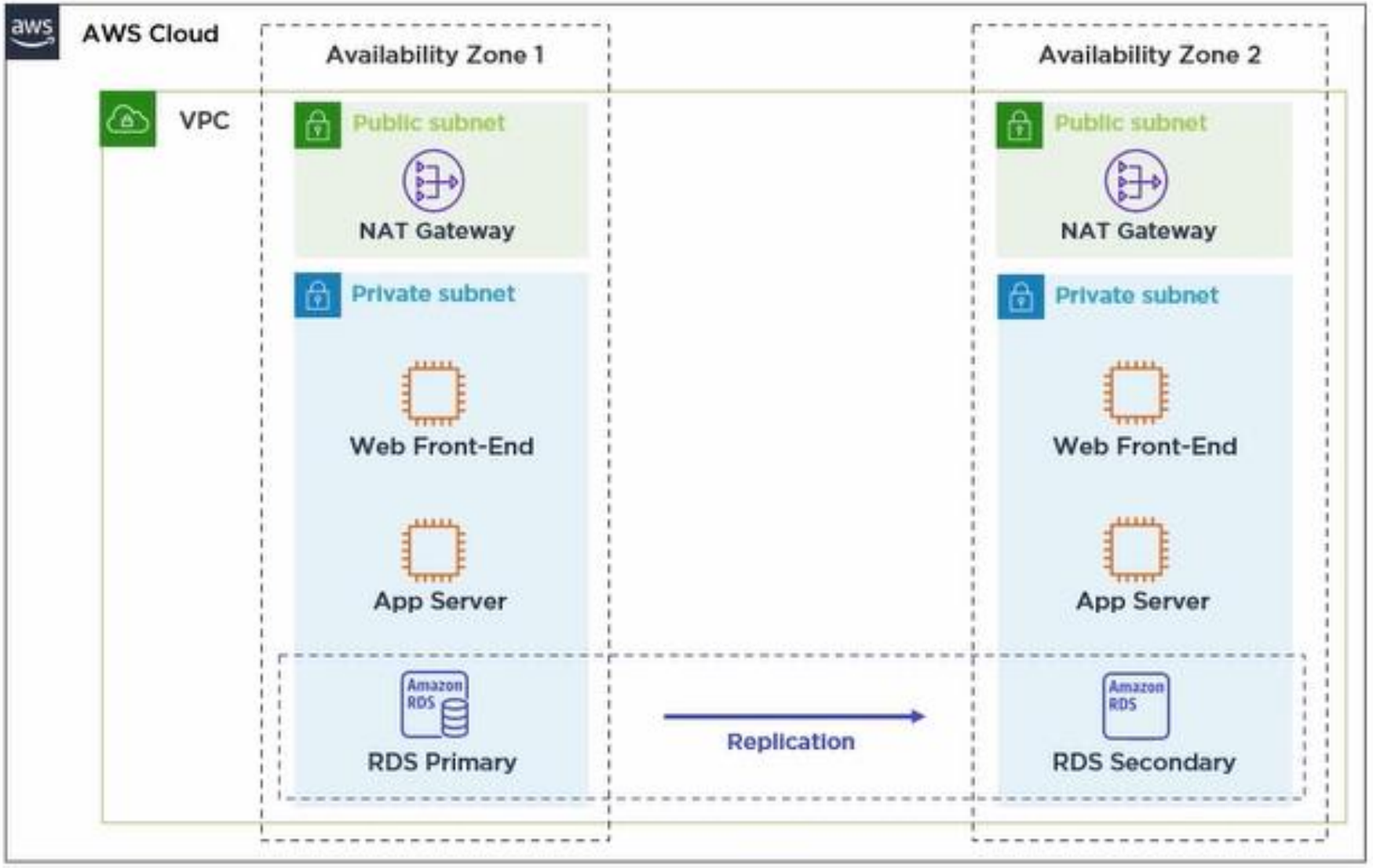
Eliminating single points of failure



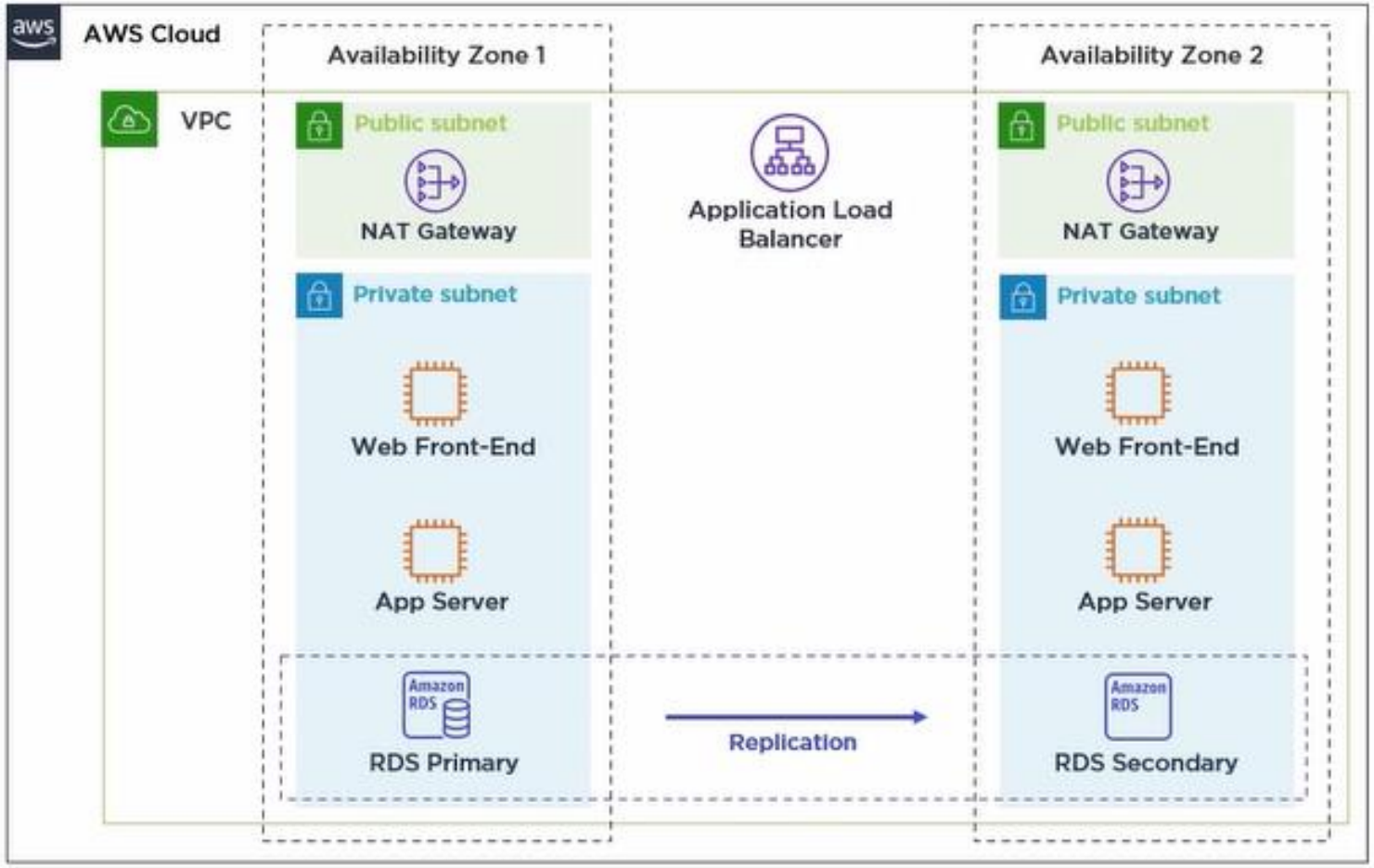
Highly Available Solution Architecture



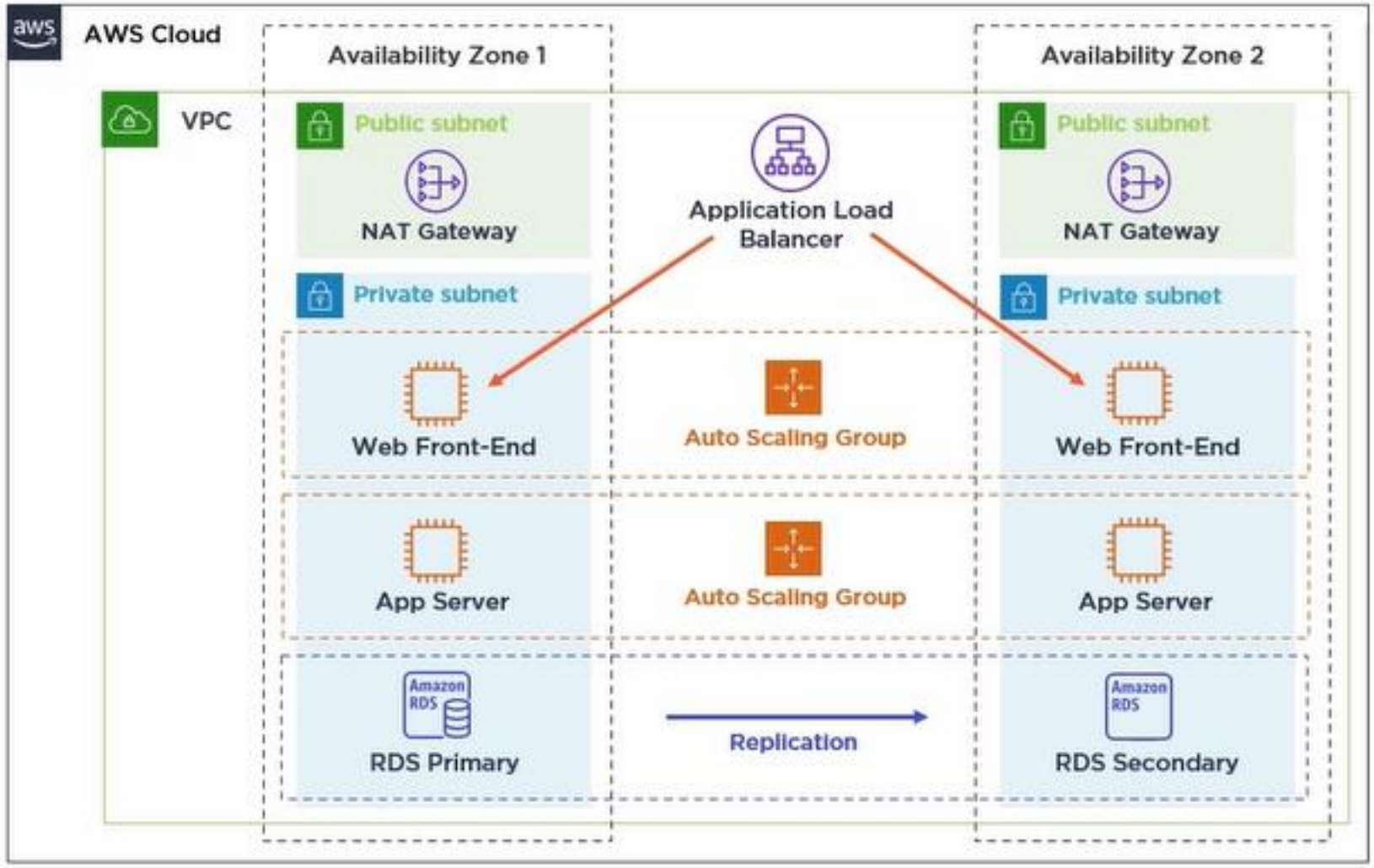
Highly Available Solution Architecture



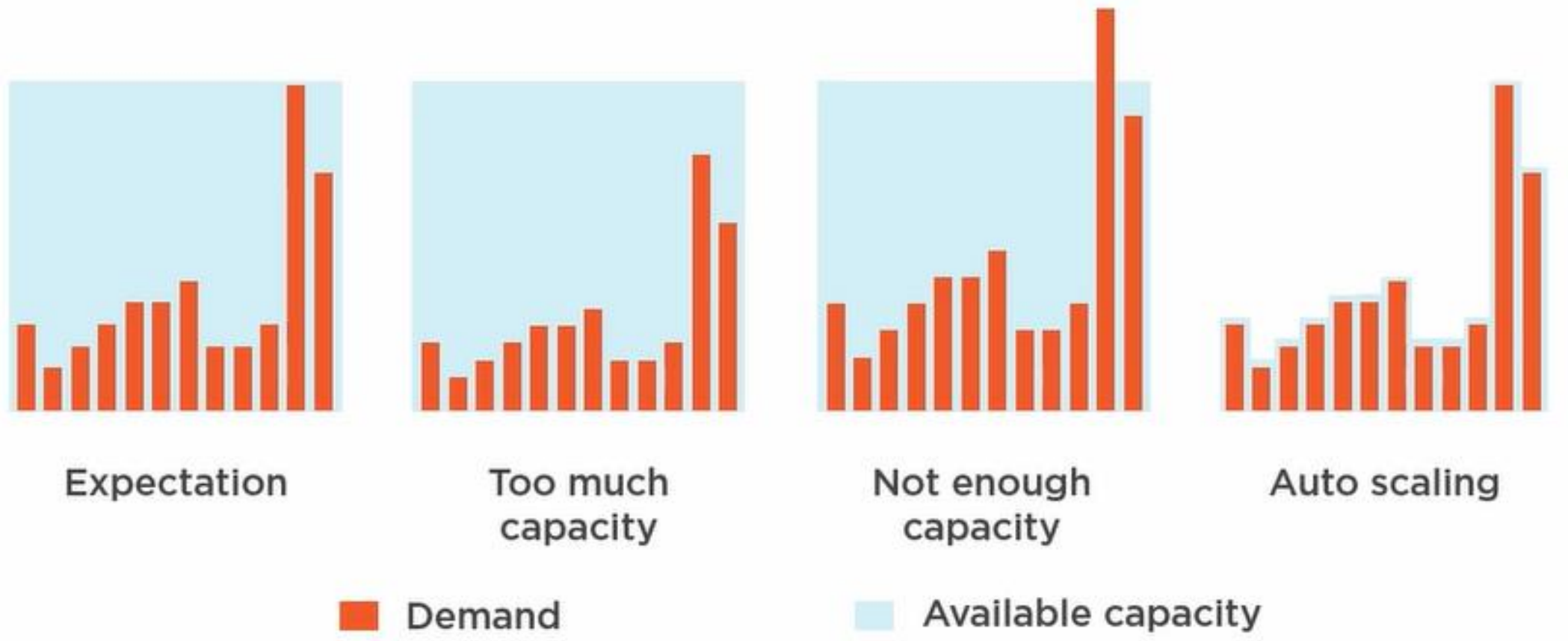
Highly Available Solution Architecture



Highly Available Solution Architecture



Capacity Planning



Horizontal and Vertical Scaling

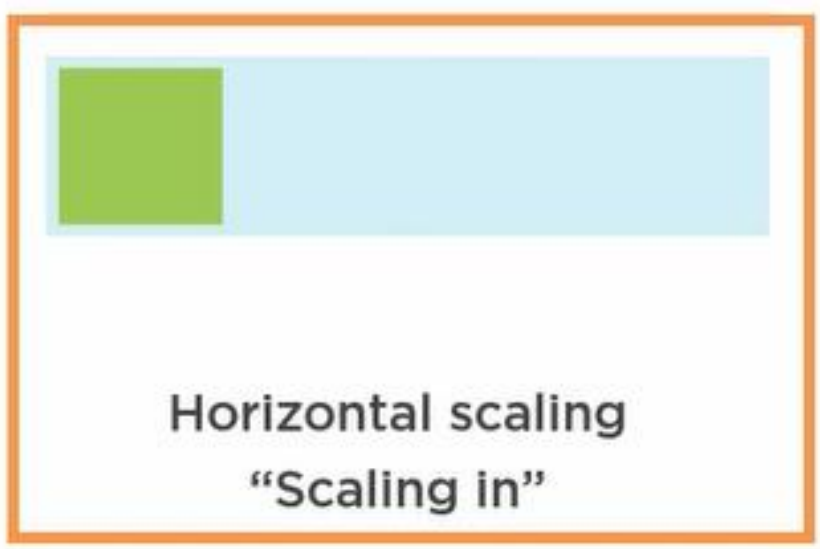


Horizontal scaling
"Scaling out"



Vertical scaling
"Scaling up"

Horizontal and Vertical Scaling



Horizontal scaling
"Scaling in"



Vertical scaling
"Scaling up"

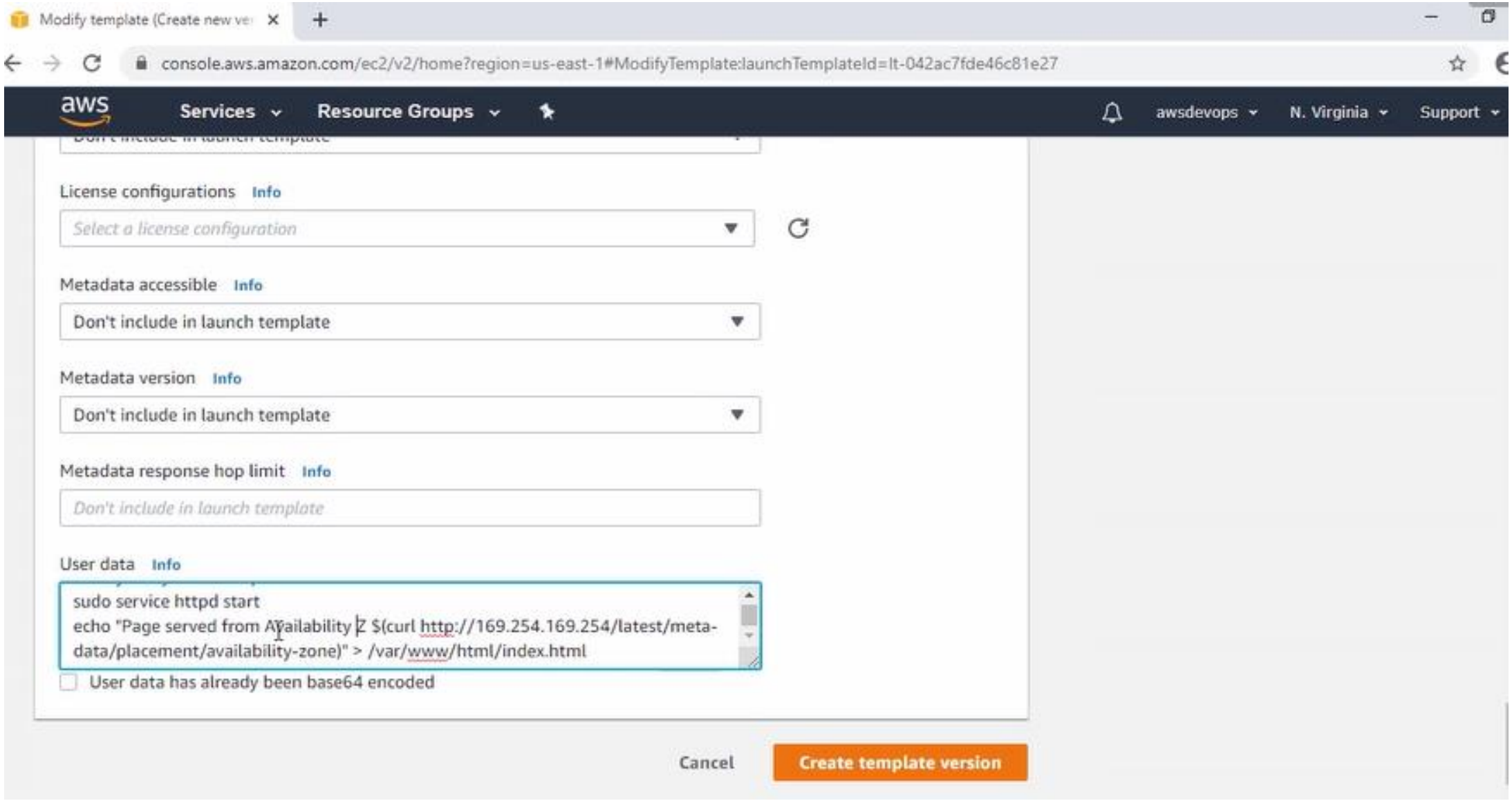
Launch Templates

Recommended by AWS over Launch Configurations

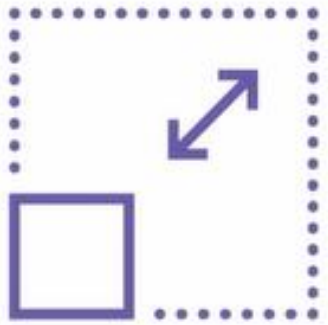
Can be changed and have multiple versions

Allow for more advanced configuration options

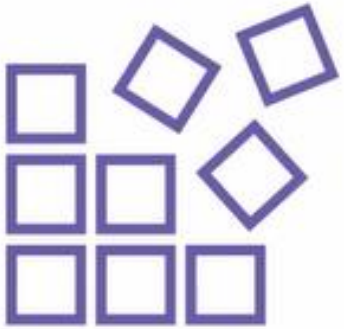
Launch Templates



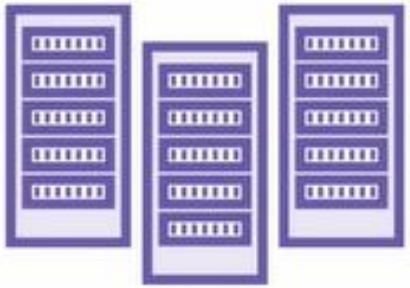
Auto Scaling



Right-sizing capacity needs



Logical collection of instances

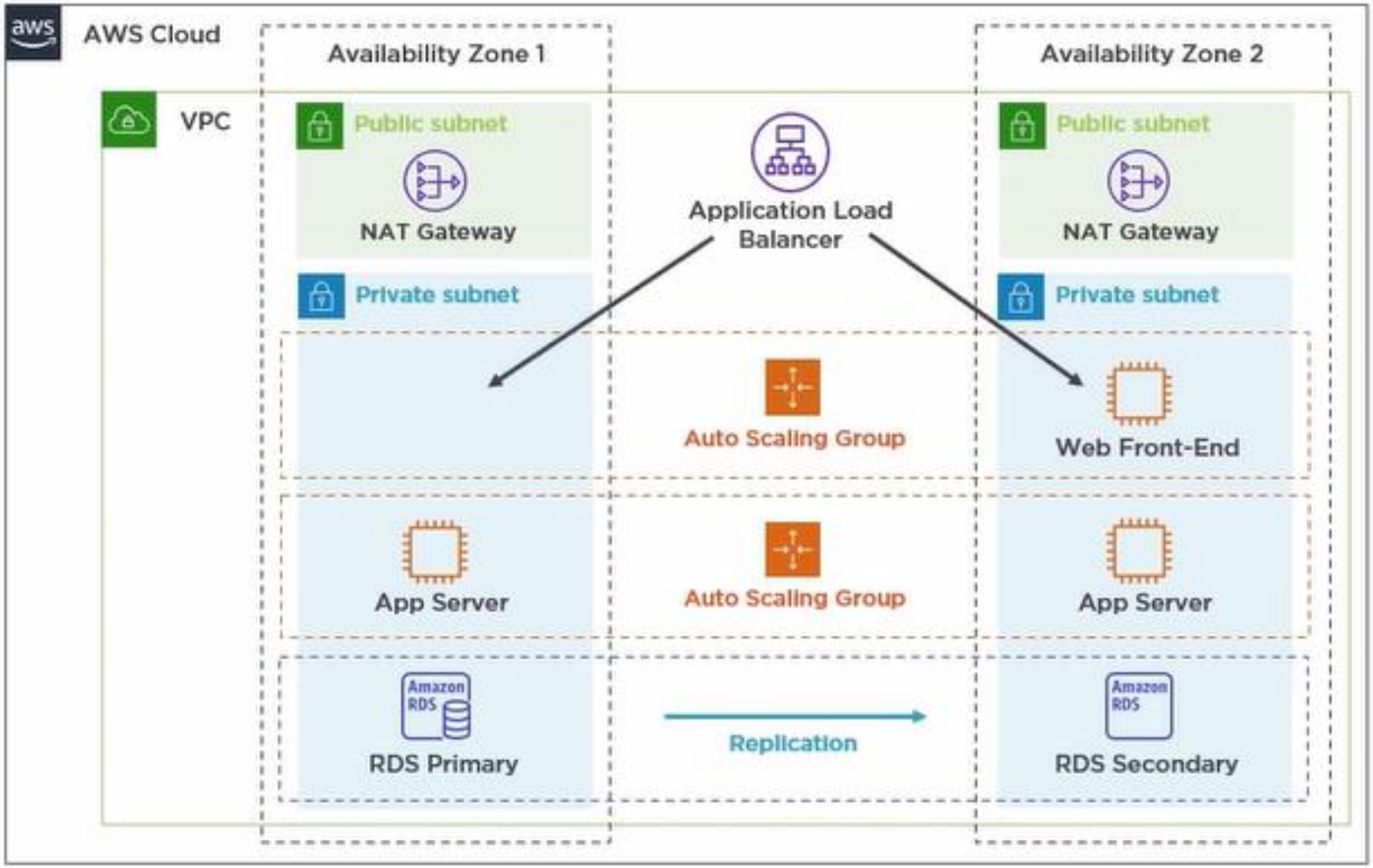


Spread across Availability Zones



Performs health checks on instances

Auto Scaling



Auto Scaling Policies

Manually adjust minimum, maximum, or desired capacity values

Automatically adjust based on metric values



Target Tracking

Auto Scaling will maintain target value



Simple

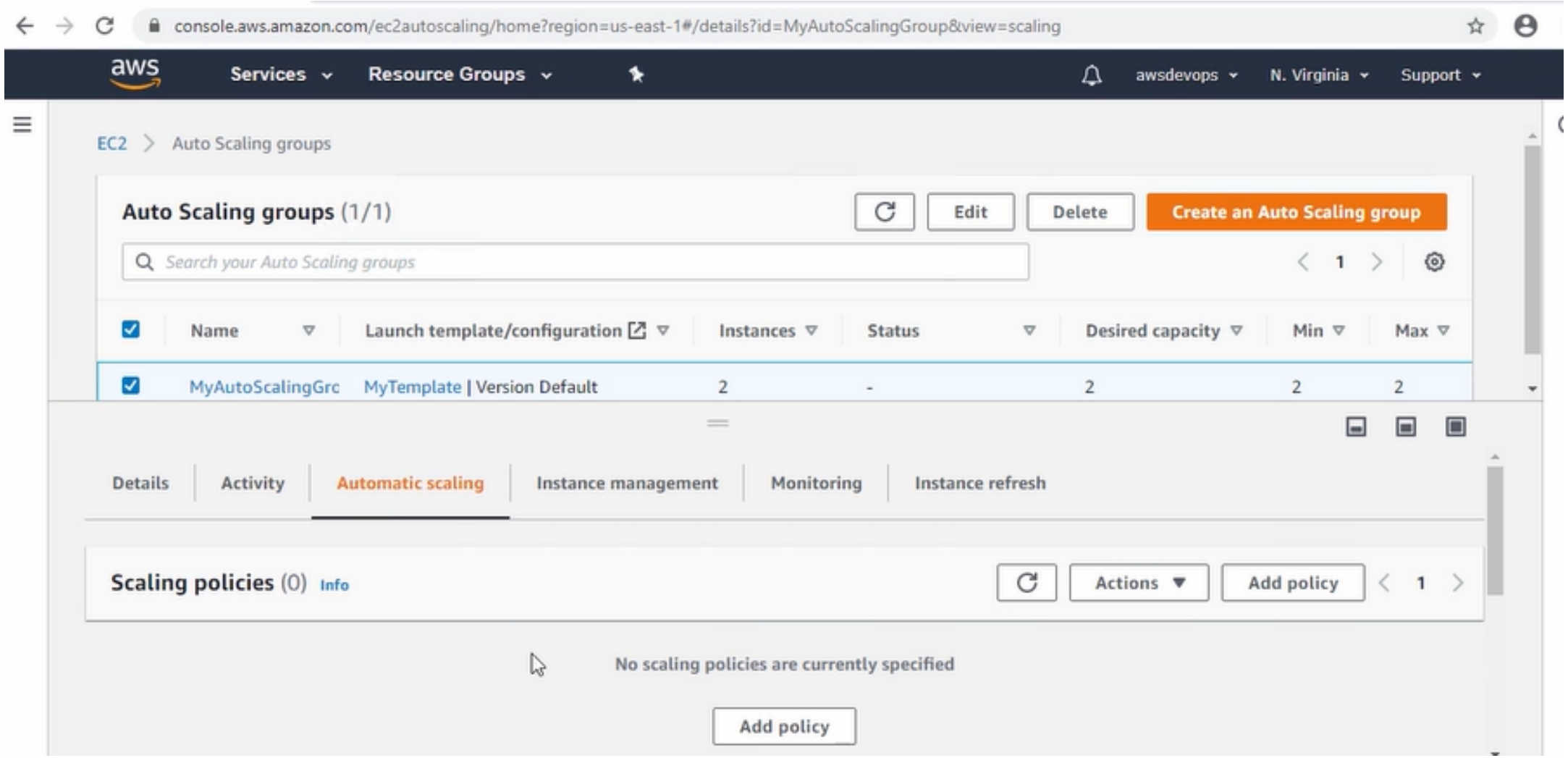
Auto Scaling acts based on CloudWatch Alarms



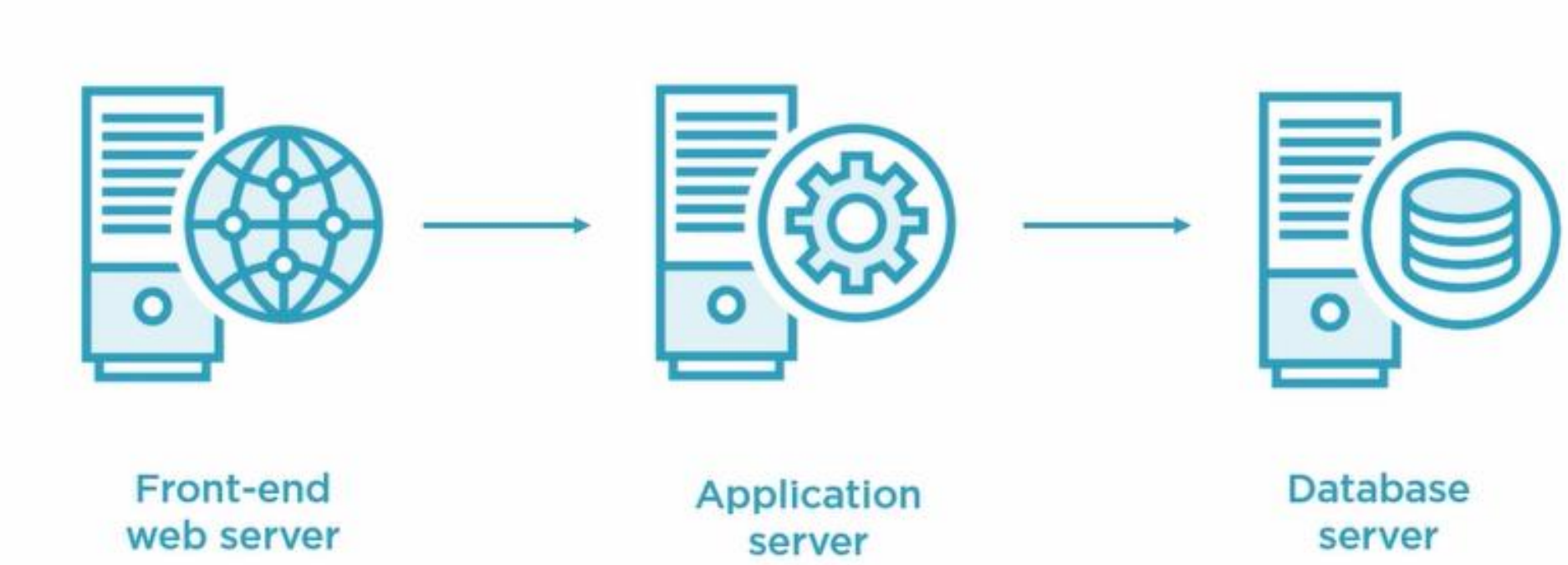
Step Scaling

Auto Scaling acts based on banded value ranges

Auto Scaling Policies



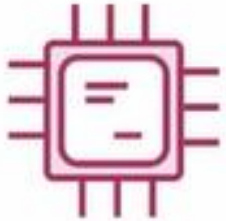
3 Tire Architecture



Amazon Relational Database Service (RDS)



Fully-managed



Leverages EC2 instances and EBS volumes



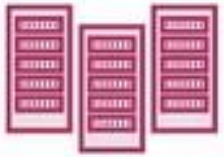
AWS handles patching and maintenance



Automated backups for point-in-time recovery



User-initiated snapshots



Multi-AZ deployments

Running Databases on EC2



Shell access to server file system or other server-side dependencies

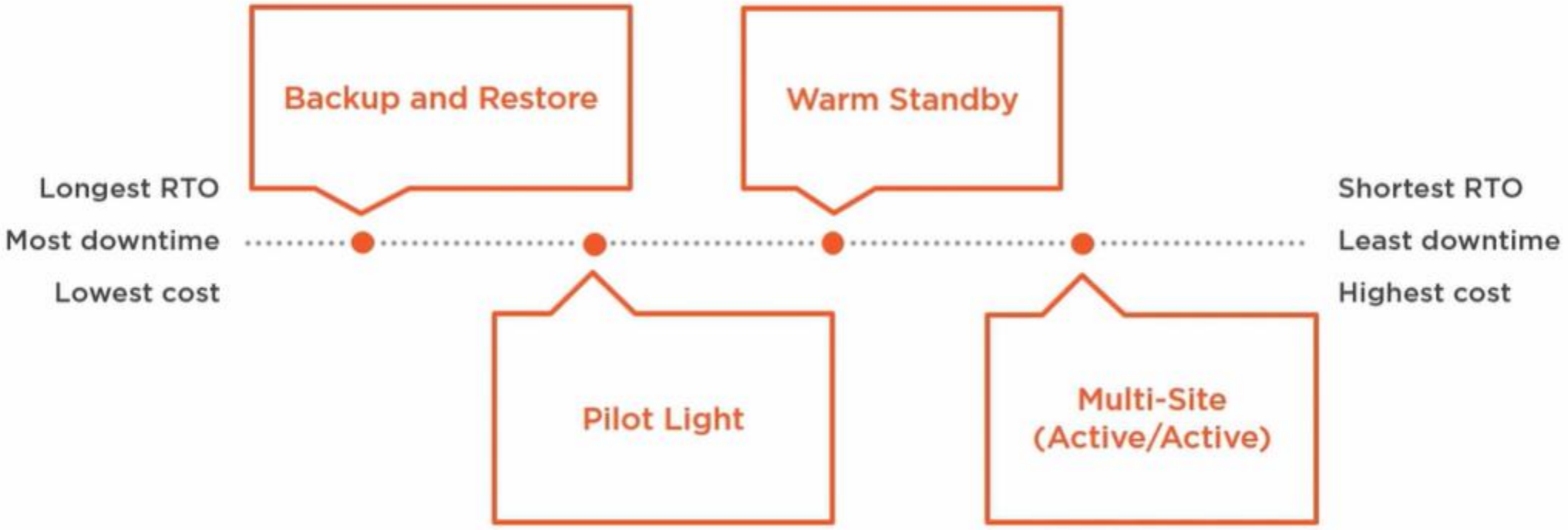


Willing to expend DBA labor in exchange for lower usage costs



Additional work required to support maintenance and failover scenarios

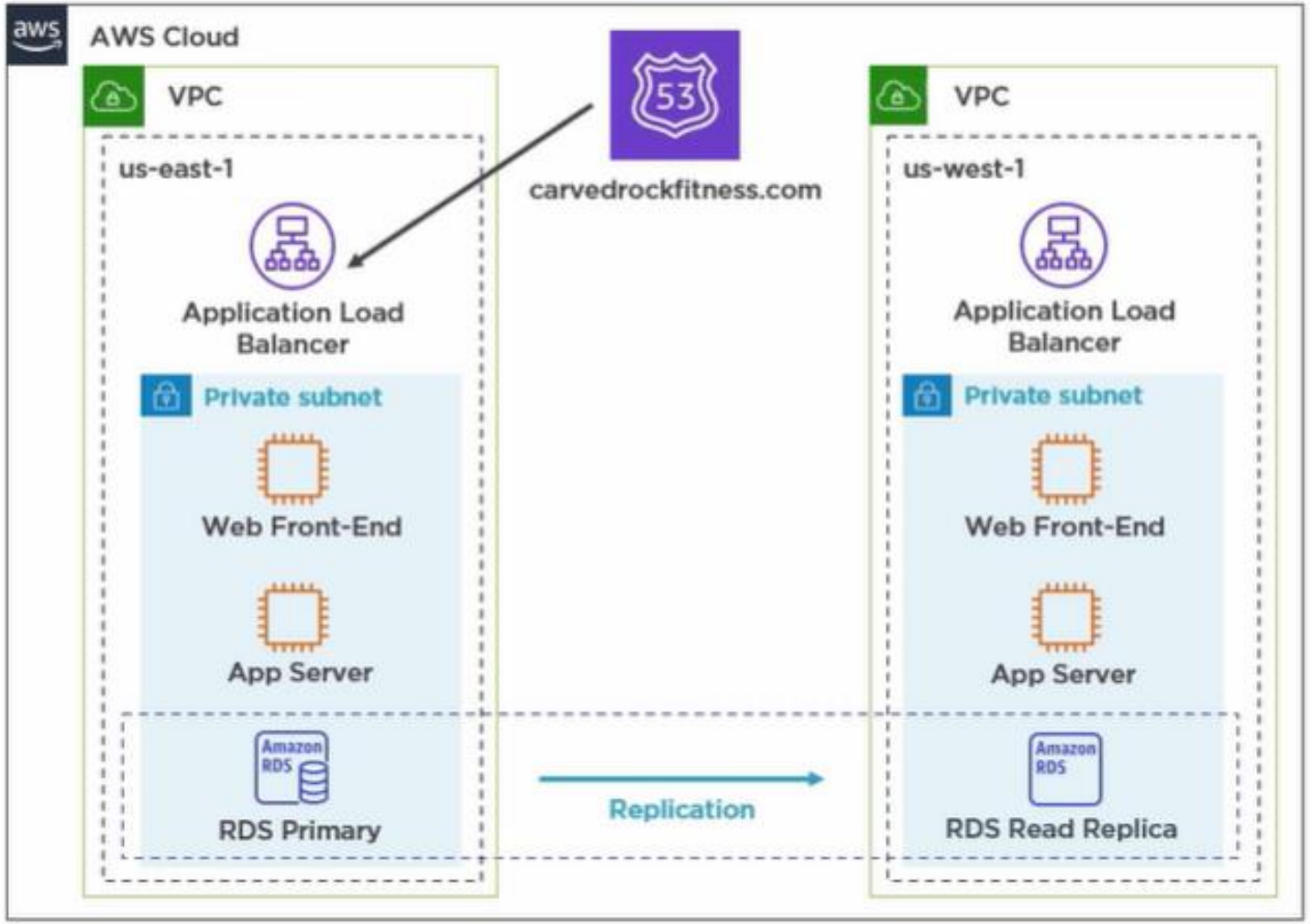
Business Continuity



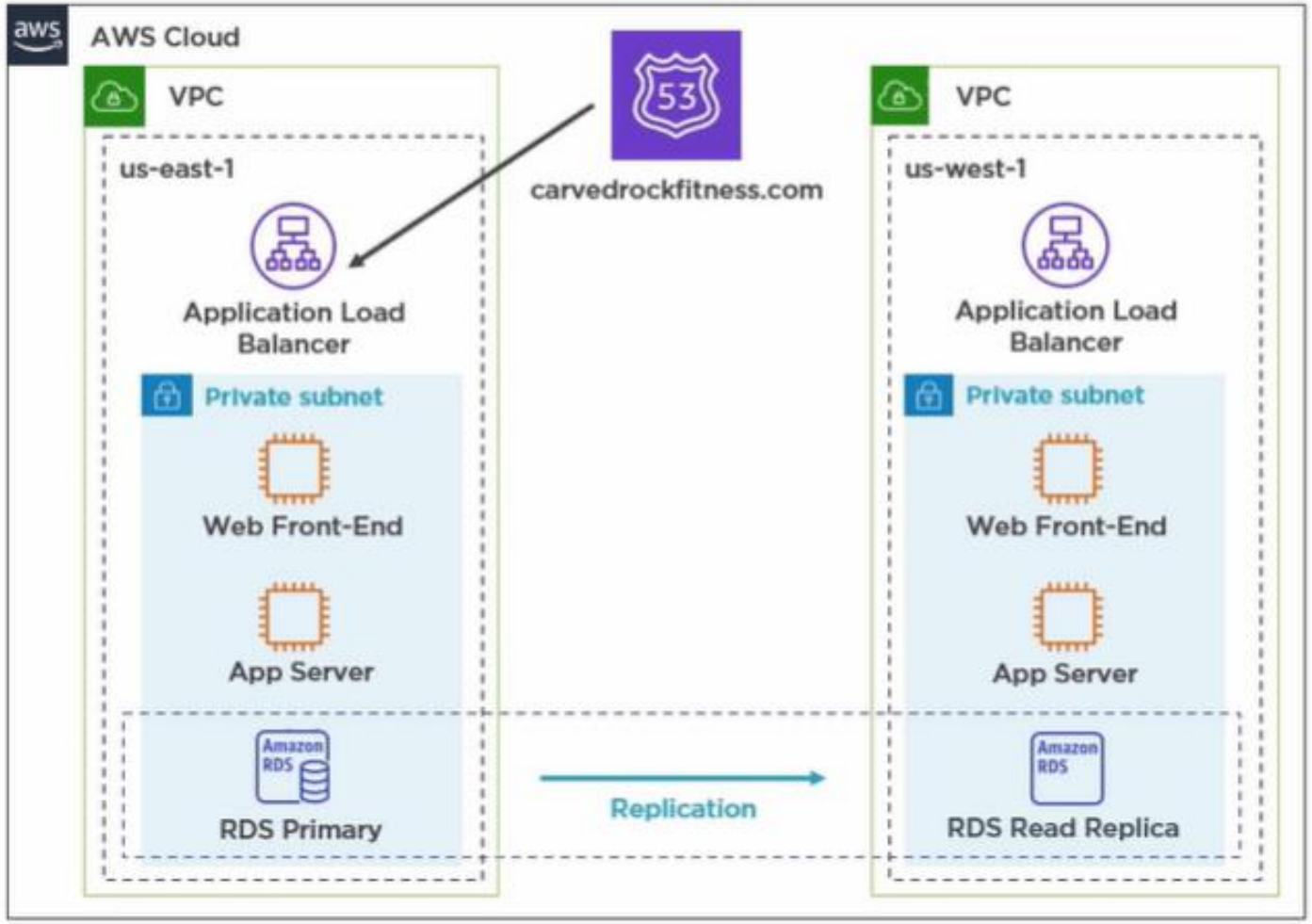
Backup and Restore



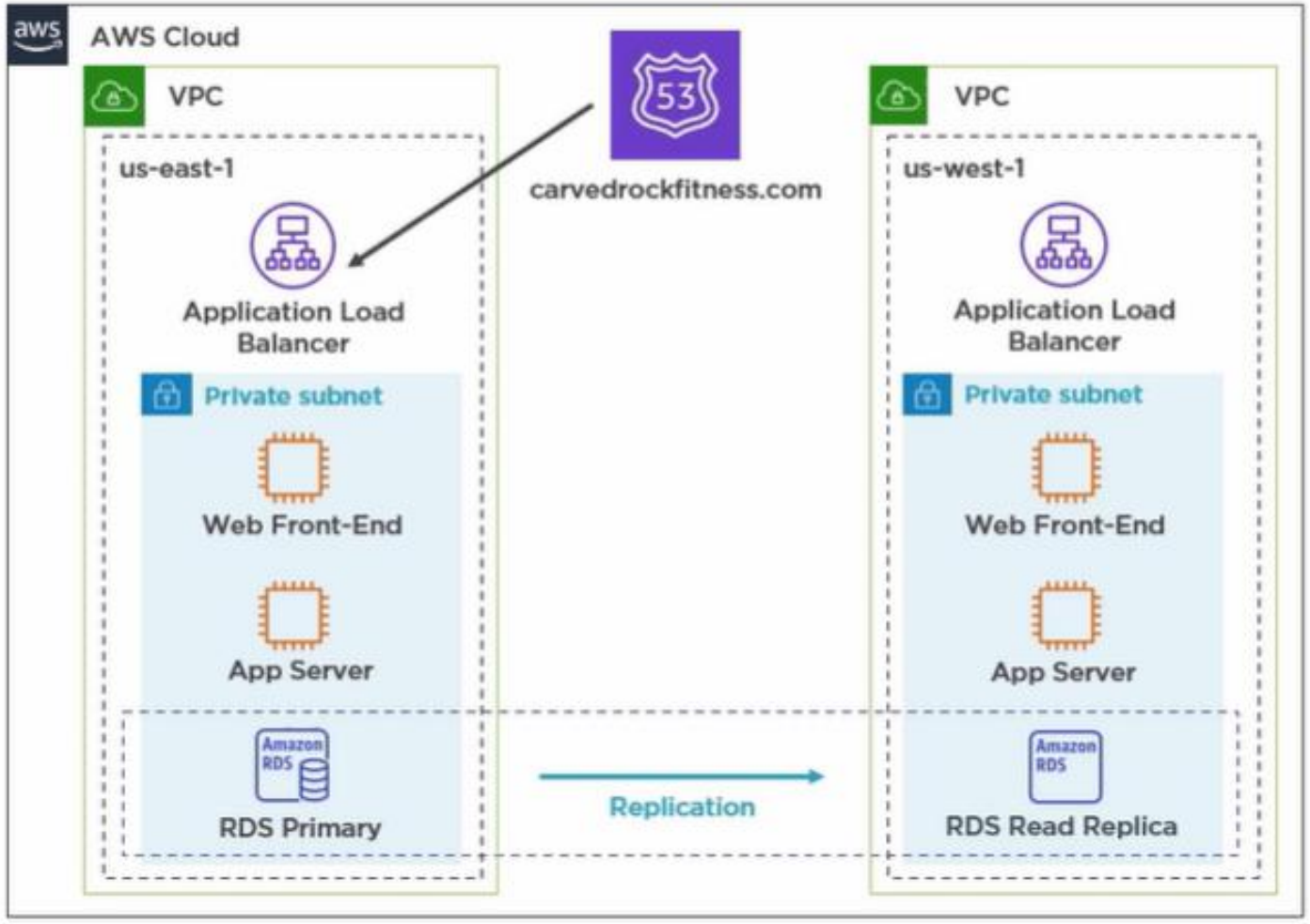
Pilot Light


AWS Cloud
VPC
us-east-1
53
carvedrockfitness.com
Application Load Balancer
Private subnet
Web Front-End
App Server
Amazon RDS
RDS Primary
Replication
VPC
us-west-1
Application Load Balancer
Private subnet
Web Front-End
App Server
Amazon RDS
RDS Read Replica

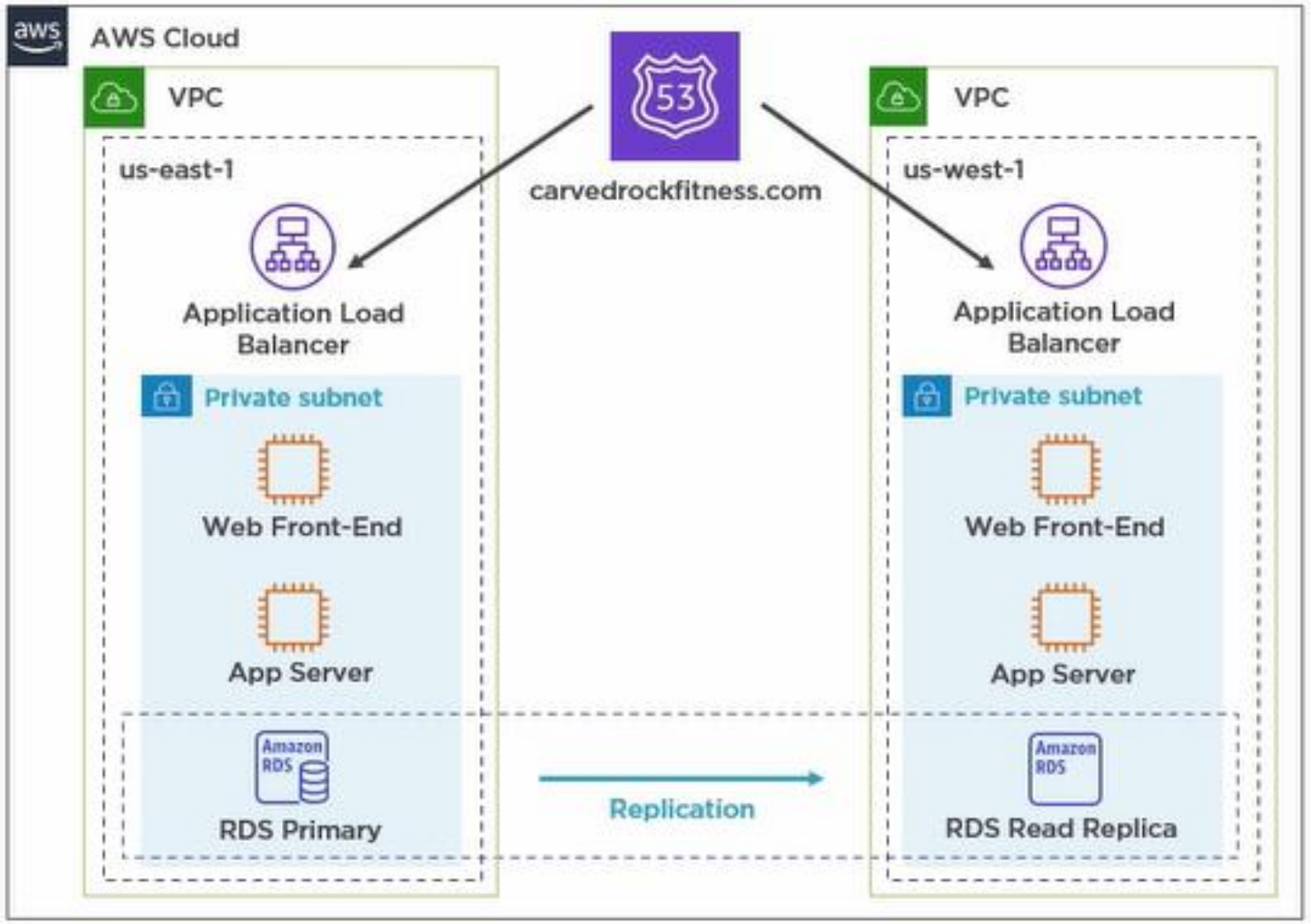
Worm Standby



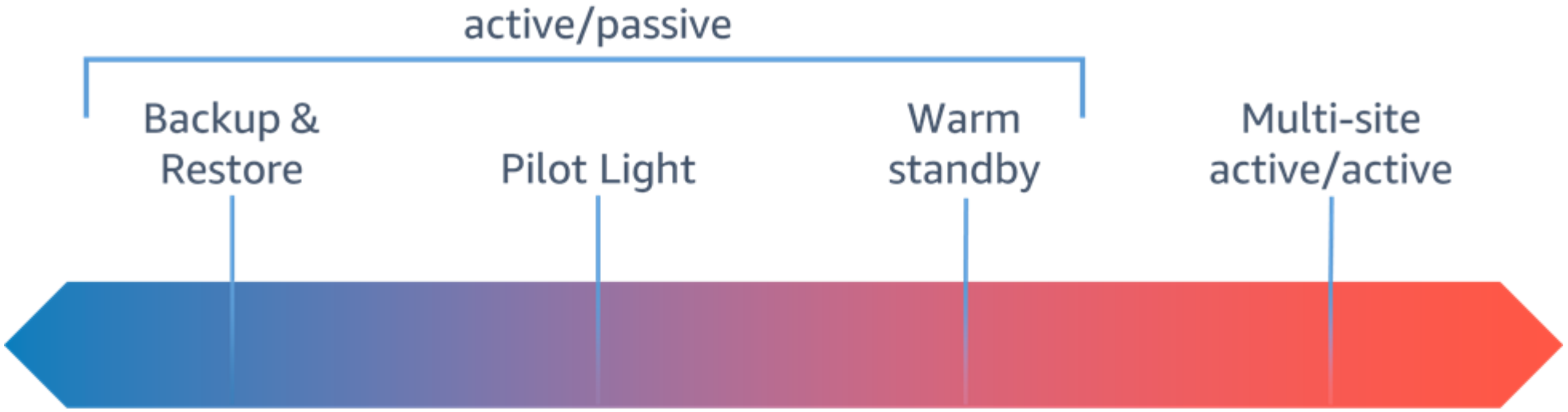
Worm Standby



Multi-Site (Active/Active)



Disaster Recovery Options



RPO / RTO: Hours

- Lower priority use cases
- Provision all AWS resources after event
- Restore backups after event
- Cost $

RPO / RTO: 10s of minutes

- Data live
- Services idle
- Provision some AWS resources and scale after event
- Cost: $$

RPO / RTO: Minutes

- Always running, but smaller
- Business critical
- Scale AWS resources after event
- Cost $$$

RPO / RTO: Real-time

- Zero downtime
- Near zero data loss
- Mission Critical Services
- Cost $$$$

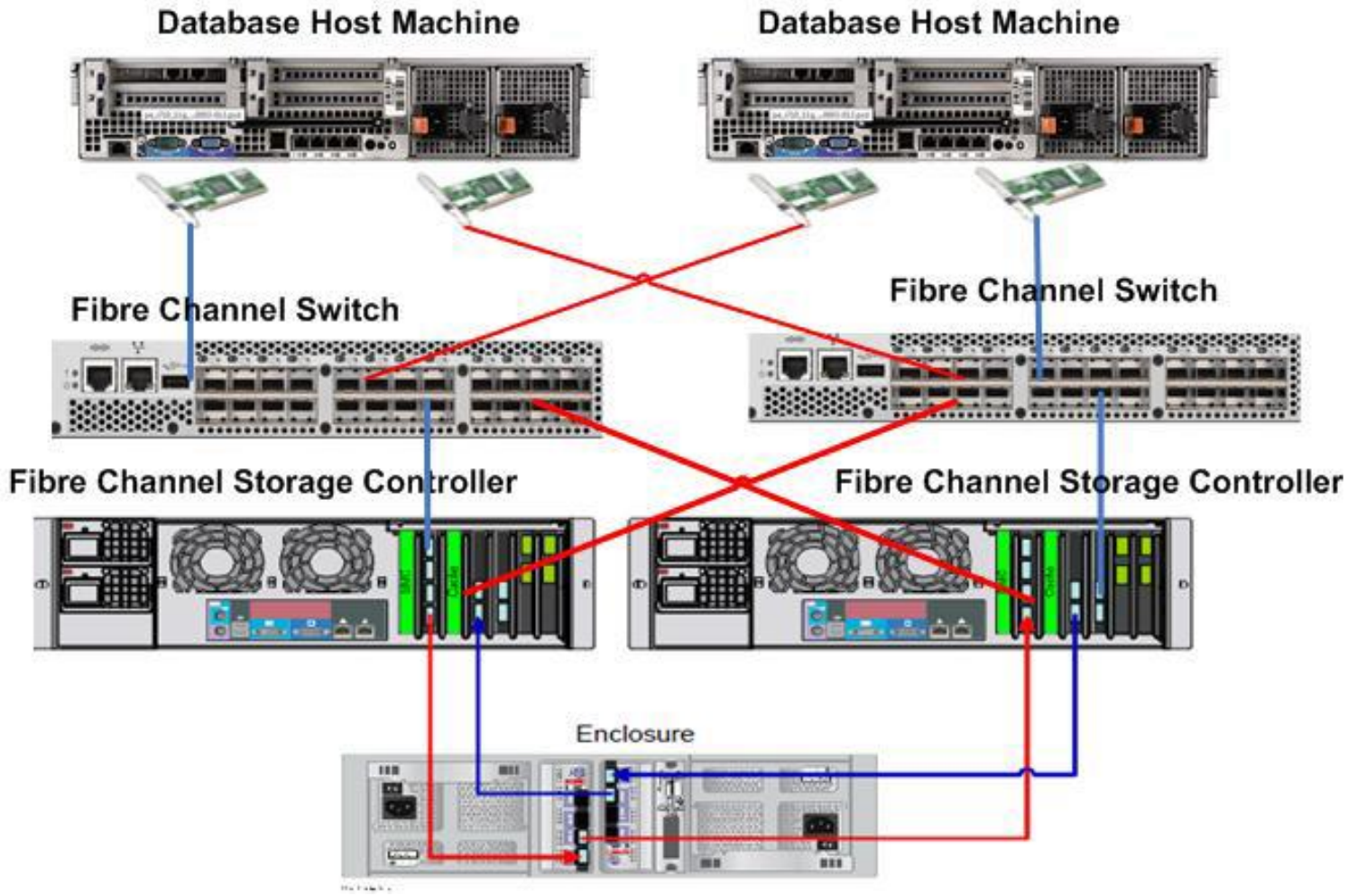
Hardware Infrastructure for High Availability

High Availability Storage Infrastructure

– Storage HA plays a key role in the infrastructure HA

– Redundant IO paths from servers to storage array Server

<-> HBAs <-> Switches <-> Storage Controllers<-> Disk Enclosures



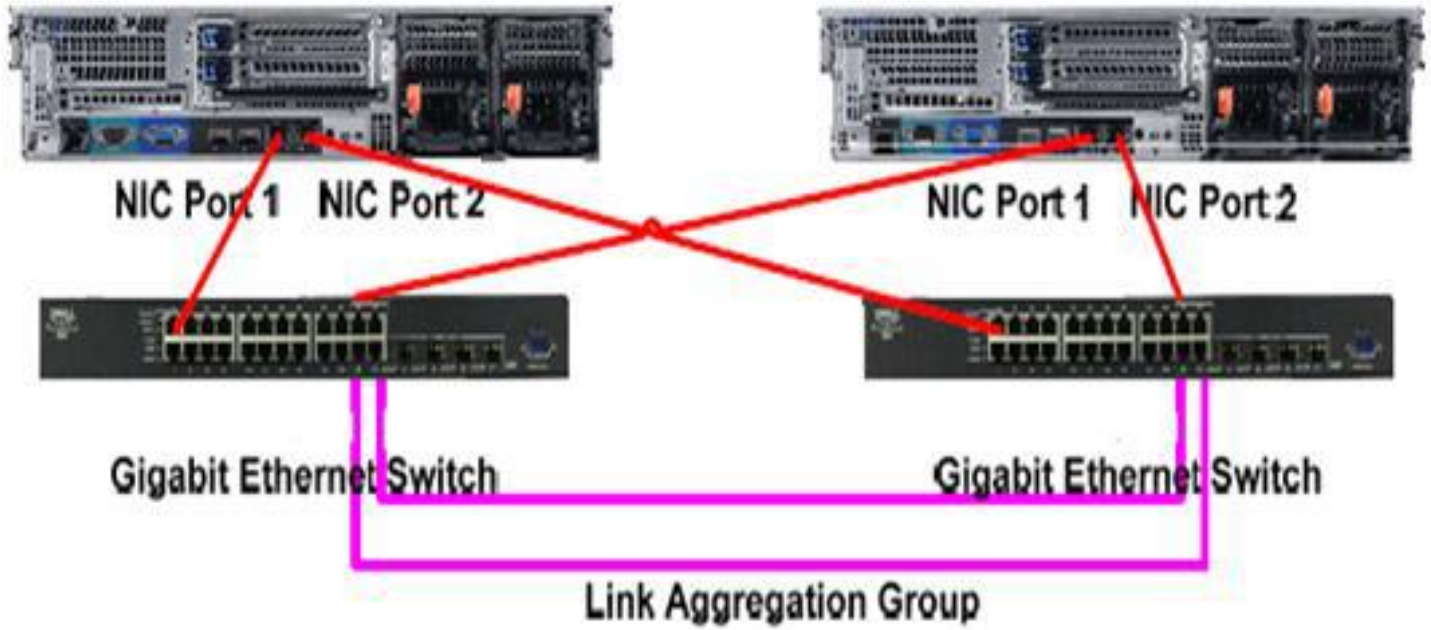
https://www.dell.com/

Hardware Infrastructure for High Availability

Network High Availability Configuration
– fully redundant interconnects for cluster configuration

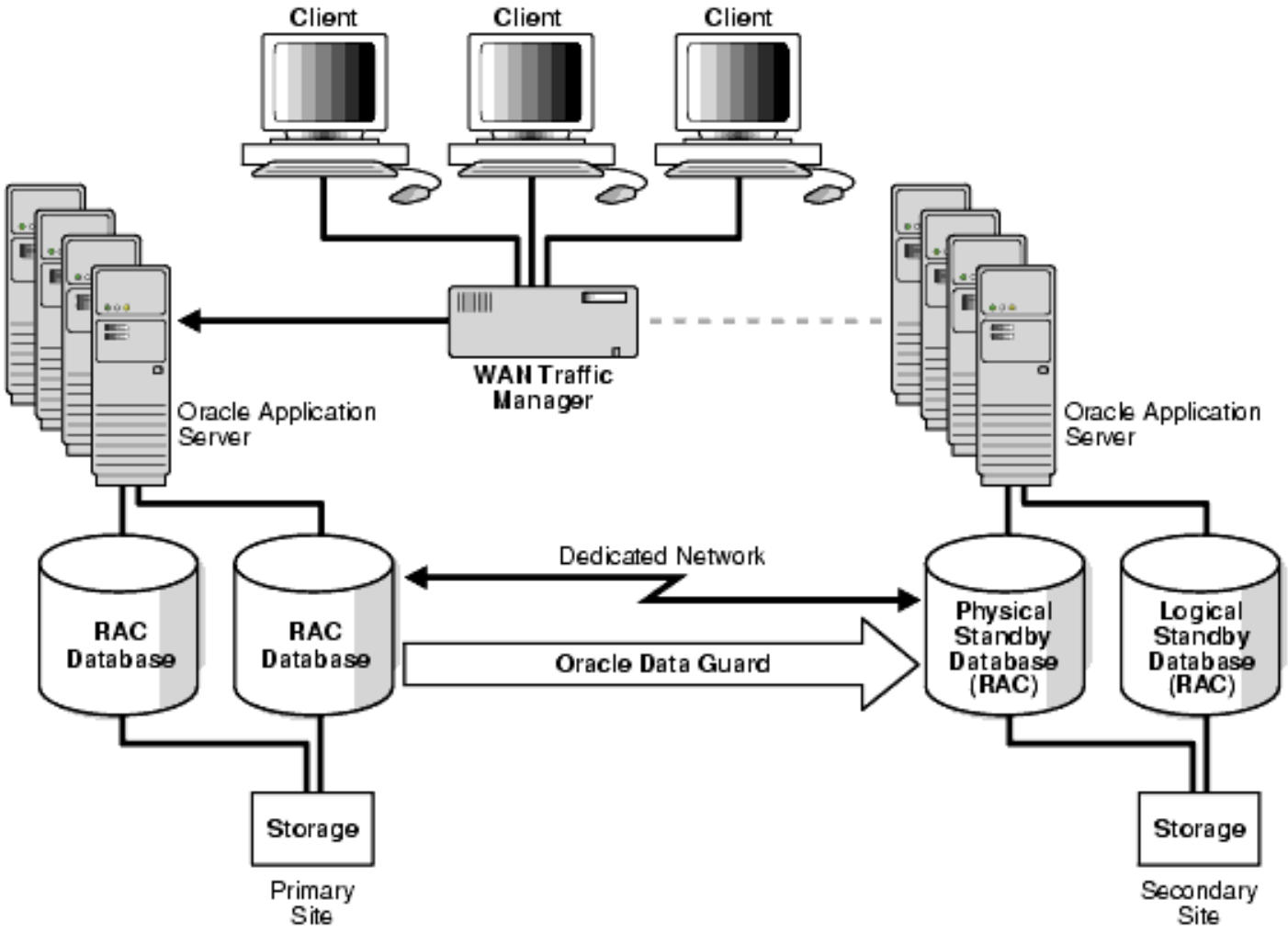
– Network bonding/teaming vs Oracle Highly Available Virtual IP (HAIP)

– Dedicated switches for private interconnects



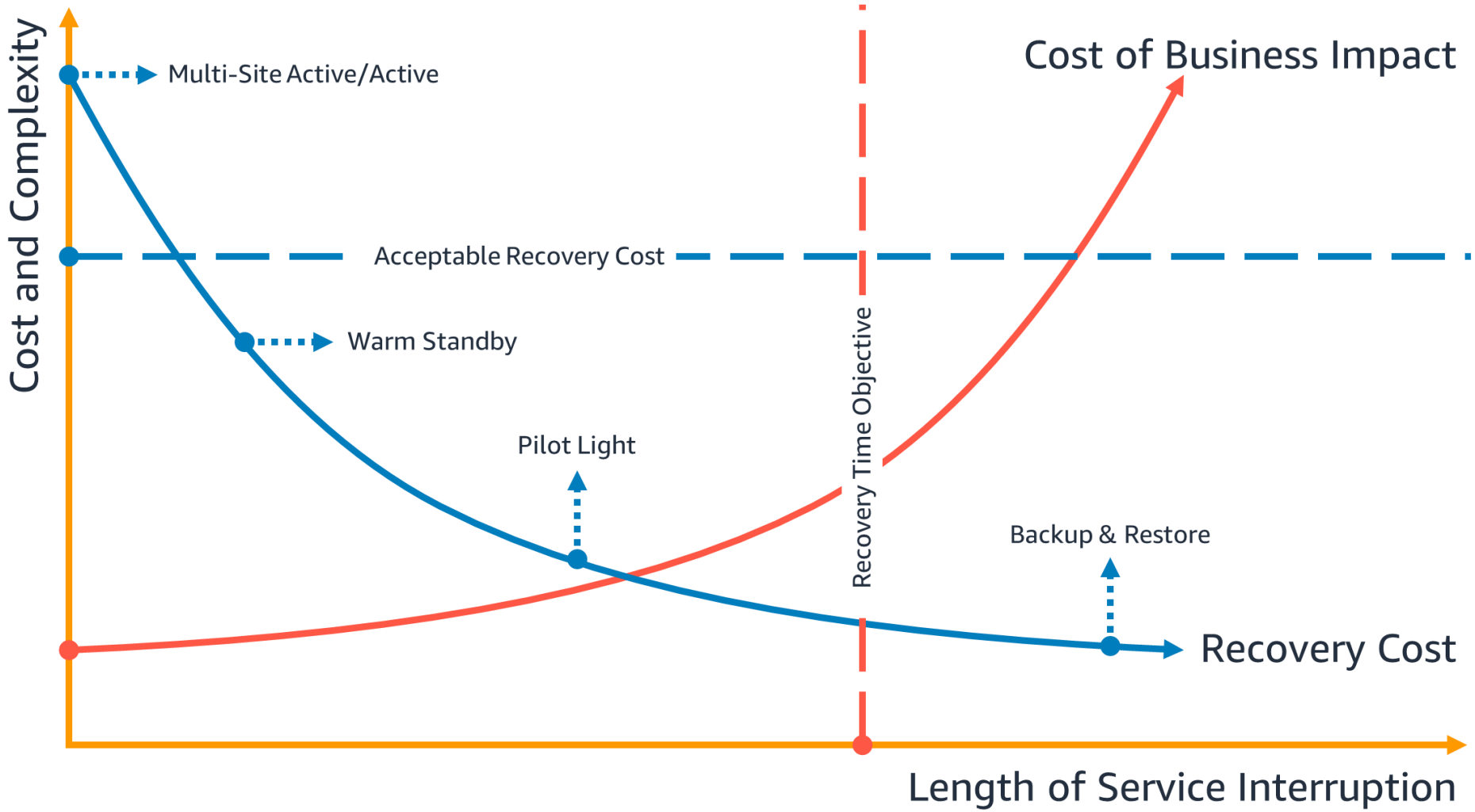
https://www.dell.com/

Oracle Cloud High availability infrastructure



https://docs.oracle.com

Oracle Cloud High availability infrastructure



https://docs.aws.amazon.com/whitepapers



Q & A



END